3

Gradient Damage Models Coupled With Plasticity and Their Application to Dynamic Fragmentation

3.1. Introduction

The initiation and propagation of cracks is still an unresolved problem in fracture mechanics. Several models have been studied in different contexts [BAR 62, BRO 94, DON 04, HAK 09], in quasi-static and dynamics [RAV 98, CHR 10] and taking into account different phenomena. The objective of this chapter is to explain the development of the so-called "gradient damage models" [AMO 11] and their extension to ductile materials and dynamic loading. The main idea of these models is that we represent the crack by a damage field and we do not need to know its path *a priori*. However, some attention must be paid to the mesh in order to avoid creating a preferential direction for crack propagation [NEG 99].

Local models have proven to be unsuitable to correctly predict damage evolution [GEE 01, PHA 11a]. Softening local damage models allow damage localization in infinitely thin bands and, consequently, cracks with zero energy dissipation [MAR 07]. In finite element simulations, this implies that the mesh size determines the size of the localization zones and the results will necessarily depend on the mesh used.

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In this context, the localization is controlled by adding a non-local term as an integral [PIJ 87, ZDE 88, BOR 96, LOR 03] or a gradient [COM 01, LOR 05, LOR 11]. The family of gradient damage models contain the gradient of damage weighed by a parameter called the characteristic length [PHA 11b] in order to avoid localization in a band of null thickness.

These models have been originally proposed for quasi-static brittle damage evolution, but have also been extended to ductile [ALE 15, AMB 15, MIE 15] and dynamic loading [BOU 11, MIC 12, LI 16b]. The main objective of this chapter is to explain the necessary changes of the original model, in order to account for plastic deformations and inertial effects.

We first present the construction of gradient damage models for brittle softening materials based on the principle of minimum energy. We discuss the main hypothesis and the need for regularization. We then briefly discuss the von Mises plasticity criterion [MAR 14] and how to take it into account [ALE 14]. We conclude the model by adding inertial effects. Once the model is complete, we will briefly discuss the numerical implementation and show a few examples and results using the *FEniCS* library [MAR 12] and an industrial code.

3.2. Theoretical aspects

3.2.1. *Gradient damage models*

We present here a simplified construction of gradient damage models for brittle elastic materials, where there are no other dissipation effects. We will consider the case of small strains theory and an isotropic material. For more details on the construction of these models, see [BOU 08, PHA 10, PHA 11b]. For the proofs of Gamma convergence, see [BRA 02, DAL 01].

We denote stress by $\boldsymbol{\sigma}$, strain by $\boldsymbol{\varepsilon}$, displacement by u and rigidity tensor by $\mathbf{E}$. The contracted product of two tensors a and b will be denoted by $a{:}b$. When working in a 1D scenario, we will denote Young's modulus by simply E, stress by σ and strain by ε.

We recall that $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla u + \nabla^T u)$ and $\boldsymbol{\sigma} = \mathbf{E}{:}\boldsymbol{\varepsilon}$

3.2.1.1. *Construction of a damage model (non-regularized)*

The objective of this section is to describe a family of damage models that can be applied to different types of materials. We will discuss the qualitative properties of these models.

We will suppose that:

1) damage can be represented by a scalar $\alpha \in [0, 1]$. When α=0, the material is healthy, and when α=1, the material is completely broken. Other choices for α are possible, for instance, $\alpha \in [0, \infty]$ or α being a tensor, but we decided to keep the model here as simple as possible;

2) the rigidity tensor $\mathbf{E}(\alpha)$ is a function of α; the material becomes less rigid when α increases and $\mathbf{E}(\alpha{=}1) = 0$ (no rigidity left when the material is broken). It is important to note that, for a fixed damage value, the stress–strain relationship is supposed to be linear;

3) damage is irreversible, i.e. it can only grow in time.

We now need to specify the conditions for damage evolution. For that, we will use an idea similar to Griffith's criterion [GRI 21], based on the notion of elastic energy restitution, in its variational form [FRA 98].

The elastic energy can be written as

$$\psi(\boldsymbol{\varepsilon}, \alpha) = \frac{1}{2}\boldsymbol{\varepsilon}{:}\mathbf{E}(\alpha){:}\boldsymbol{\varepsilon} \qquad [3.1]$$

For a fixed deformation, a small increase ($\delta\alpha > 0$) of damage causes a loss of $-\frac{\partial\psi}{\partial\alpha}(\boldsymbol{\varepsilon}, \alpha)\delta\alpha{>}0$ in the elastic energy. We compare the variation of elastic energy to a threshold $k(\alpha)$. As in Griffith's model, the rate of energy restitution is always lower than or equal to a threshold value and the crack only propagates when we have an equality. For this family of damage models, the propagation criterion can be written as

$$-\frac{1}{2}\boldsymbol{\varepsilon}{:}\mathbf{E}'(\alpha){:}\boldsymbol{\varepsilon} \leq k(\alpha), \qquad \begin{cases} \dot\alpha = 0 & \text{if } -\frac{1}{2}\boldsymbol{\varepsilon}{:}\mathbf{E}'(\alpha){:}\boldsymbol{\varepsilon} < k(\alpha) \\ \dot\alpha \geq 0 & \text{if } -\frac{1}{2}\boldsymbol{\varepsilon}{:}\mathbf{E}'(\alpha){:}\boldsymbol{\varepsilon} = k(\alpha) \end{cases} \qquad [3.2]$$

where $k(\alpha)$ is a function of α representing the energy restitution necessary for damage to evolve.

So far, we have made only a few hypotheses concerning $\mathbf{E}(\alpha)$ and $k(\alpha)$. One of the strengths of such models is that the specific forms of $\mathbf{E}(\alpha)$ can be chosen so as to describe the behavior of a specific material.

EXAMPLE 3.1.– *As a first example, we will consider a 1D bar under traction, where damage increases uniformly in space. We consider the functions*

$$E(\alpha) = E_0(1 - \alpha)^2 \quad \textit{and} \quad k(\alpha) = k_0, \qquad [3.3]$$

where k_0 is a constant and E_0 is Young's modulus when the material has not yet suffered any damage.

It is easy to see that the damage criterion can be written as

$$(1-\alpha)E_0\varepsilon^2 \le k_0, \qquad \begin{cases} \dot{\alpha} = 0 & \textit{if } (1-\alpha)E_0\varepsilon^2 < k_0 \\ \dot{\alpha} \ge 0 & \textit{if } (1-\alpha)E_0\varepsilon^2 = k_0 \end{cases}$$

If the initial bar is not damaged, then the bar will not be damaged while $E_0\varepsilon^2 < k_0$. We can define critical strain ε_c and critical stress σ_c by

$$\varepsilon_c = \sqrt{\frac{k_0}{E_0}} \quad \textit{and} \quad \sigma_c = E_0\varepsilon_c = \sqrt{k_0 E_0}$$

When $\varepsilon > \varepsilon_c$, we can use the criterion once again to find that $(1-\alpha)E_0\varepsilon^2 = k_0$ and thus

$$\alpha = 1 - \frac{k_0}{E_0\varepsilon^2}$$

For this model, we can see that while $\varepsilon \le \varepsilon_c$, stress linearly increases with strain and the bar suffers no damage. When $\varepsilon > \varepsilon_c$, the bar damages, which causes a stress–strain relationship that is no longer linear.

Figure 3.1 shows the normalized stress $\bar{\sigma} = \sigma/\sigma_c$ and damage α as a function of the normalized strain $\bar{\varepsilon} = \varepsilon/\varepsilon_c$.

Let $w(\alpha)$ be a function such that $w'(\alpha) = k(\alpha)$. We define the energy density by

$$W(\boldsymbol{\varepsilon}, \alpha) = \psi(\boldsymbol{\varepsilon}, \alpha) + w(\alpha) \qquad [3.4]$$

We can write the stress as

$$\boldsymbol{\sigma} = \frac{\partial W}{\partial \boldsymbol{\varepsilon}}(\boldsymbol{\varepsilon}, \alpha) \qquad [3.5]$$

and the damage evolution criterion as

$$\frac{\partial W}{\partial \alpha}(\boldsymbol{\varepsilon}, \alpha) \cdot \dot{\alpha} = 0, \qquad [3.6]$$

where each of the two factors is non-negative.

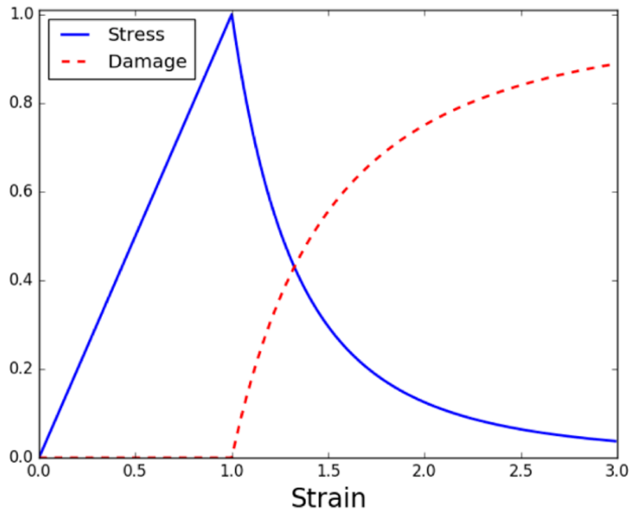

Figure 3.1. *Damage (dashed red line) and normalized stress (blue line) for a 1D bar subject to traction, according to the model given in example 3.1. For a color version of the figure, see www.iste.co.uk/lambert/dynamic.zip*

Now let $\dot{\beta} \geq 0$ be a small increase of damage in time. We have that

$$\frac{\partial W}{\partial \alpha}(\boldsymbol{\varepsilon}, \alpha)\cdot(\dot{\beta} - \dot{\alpha}) \geq 0 \qquad [3.7]$$

Consider a structure whose initial configuration is given by $\Omega \subset \mathbb{R}^n$ $(n{=}1, 2, 3)$.

Suppose we have a volume force f acting on the whole structure, an imposed displacement u_0 on $\partial_u \subset \partial\Omega$ and a normal stress T on $\partial_T \subset \partial\Omega$. We also suppose that $\partial_u \bigcap \partial_T = \emptyset$ and $\partial_u \bigcup \partial_T = \partial\Omega$. The static equilibrium can be written as

$$\begin{cases} \mathrm{div}\boldsymbol{\sigma} + f = 0 \quad \text{on } \Omega \\ u = u_D \quad \text{on } \partial_u \\ \boldsymbol{\sigma}\cdot n = T \quad \text{on } \partial_T \end{cases} \qquad [3.8]$$

We fix a test function w such that $w = 0$ on ∂_u. Then

$$\int_\Omega \big(\mathrm{div}\boldsymbol{\sigma}\cdot w + f\cdot w\big)d\Omega = 0 \qquad [3.9]$$

and Green's formula shows that

$$\underbrace{\int_{\partial u}(\boldsymbol{\sigma}\cdot n)\cdot wdS}_{0} + \int_{\partial T} T\cdot wdS - \int_\Omega \boldsymbol{\sigma}:\boldsymbol{\varepsilon}(w)d\Omega + \int_\Omega f\cdot wd\Omega = 0. \qquad [3.10]$$

We define

$$\begin{aligned} \mathcal{C} &= \{u : u{=}u_D \text{ on } \partial_u\} \\ \mathcal{C}_0 &= \{w : w{=}0 \text{ on } \partial_u\} \end{aligned} \qquad [3.11]$$

The static equilibrium problem consists of finding $u \in \mathcal{C}$ such that

$$\int_\Omega \frac{\partial W}{\partial \boldsymbol{\varepsilon}}(\boldsymbol{\varepsilon}(u), \alpha):\boldsymbol{\varepsilon}(w)d\Omega = \int_\Omega f\cdot wd\Omega + \int_{\partial T} T\cdot wdS, \quad \forall w \in \mathcal{C}_0. \qquad [3.12]$$

If we consider the evolution problem, where the time is denoted by t, by integrating [3.7], we obtain the following problem: find $\dot\alpha \geq 0$ such that

$$\int_\Omega \frac{\partial W}{\partial \alpha}(\boldsymbol{\varepsilon}, \alpha)\cdot(\dot\beta - \dot\alpha)d\Omega \geq 0, \quad \forall \dot\beta \geq 0 \qquad [3.13]$$

We define the total energy of the system by

$$\mathcal{E}(u, \alpha) = \int_\Omega W(\boldsymbol{\varepsilon}(u), \alpha)d\Omega - \int_\Omega f\cdot ud\Omega - \int_{\partial T} T\cdot udS \qquad [3.14]$$

It is easy to see that solving these two equations is equivalent to finding a local minimum for the energy $\mathcal{E}$. We denote the total derivative of $\mathcal{E}$ by $D\mathcal{E}$.

The evolution problem, given by equations [3.12] and [3.13], is equivalent to finding $u \in \mathcal{C}$ and $\dot\alpha{\geq}0$ such that for all $v \in \mathcal{C}$ and $\dot\beta{\geq}0$, we have

$$D\mathcal{E}(u, \alpha)(v - u, \dot\beta - \dot\alpha) \geq 0, \forall v \in \mathcal{C}, \forall \dot\beta \geq 0 \qquad [3.15]$$

3.2.1.2. *Regularized model*

It is now a well-known fact that local softening damage models are not viable [ALE 15, AMO 11] as they allow damage localization in infinitely thin bands.

EXAMPLE 3.2.– *To illustrate the problem of damage localization, we will consider a 1D bar of length L and a material such that $E(\alpha) = E_0(1-\alpha)^2$ and $w(\alpha) = w_1\alpha$.*

When in equilibrium, we know that $\sigma(x) = \sigma$ (constant).

We will show that for any $0 < \theta < 1$ fixed, we can construct a solution to the damage problem such that there is no damage in the interval $(0, \theta L)$ and uniform damage in $(\theta L, L)$.

In fact, for $x \in (0, \theta L)$, we have $\varepsilon(x) = \sigma/E_0$.

For $x \in (\theta L, L)$, the damage criterion can be written as

$$w_1 = -\frac{1}{2}E'(\alpha)\frac{\sigma^2}{E(\alpha)^2} = E_0(1-\alpha)\frac{\sigma^2}{E_0^2(1-\alpha)^4} = \frac{\sigma^2}{E_0}\frac{1}{(1-\alpha)^3} \qquad [3.16]$$

Therefore, damage in this interval is given by

$$\alpha^* = 1 - \sqrt[3]{\frac{\sigma^2}{w_1 E_0}} \qquad [3.17]$$

The dissipated energy can be calculated as

$$\mathcal{D} = \int_0^L w(\alpha)dx = \int_{\theta L}^L w_1\alpha^* dx = w_1\alpha^*(1-\theta)L \qquad [3.18]$$

This shows that we have a solution to the damage problem for any θ. We can see that damage can be localized in an infinitely thin band and if we take $\theta \to 1$, the dissipated energy $\mathcal{D}$ tends to zero.

In a finite element code, the size of the damage band will be determined by the mesh size. This means that refining the mesh will produce different results and dissipated energies that can tend to zero.

To solve this problem, a regularizing term is used. The main idea is to add a *characteristic length* in order to penalize sharp damage profiles and solve the problem of localization in thin bands and fracture without energy dissipation.

One simple way of doing this is by adding a term that depends on the norm of the gradient of damage.

We will use an energy density of the form

$$W(\boldsymbol{\varepsilon}, \alpha, \nabla\alpha) = \psi(\boldsymbol{\varepsilon}, \alpha) + w(\alpha) + \frac{1}{2} w_1 \ell^2 \nabla\alpha \cdot \nabla\alpha, \qquad [3.19]$$

where ℓ is the characteristic length and $w_1 > 0$ is a normalization constant.

In the previous section, when describing the model, we first proposed an evolution law based on the energy restitution rate. We then expressed the static equilibrium and damage evolution by a principle of minimum energy. For this new energy density, we will directly use the principle of minimum energy to obtain an evolution law, instead of manually proposing it.

We have

$$\boldsymbol{\sigma} = \mathbf{E}(\alpha):\boldsymbol{\varepsilon} = \frac{\partial W}{\partial \boldsymbol{\varepsilon}}(\boldsymbol{\varepsilon}, \alpha) \qquad [3.20]$$

We define the dissipated energy by

$$\mathcal{D}(\alpha) = \int_\Omega w(\alpha) + \frac{1}{2} w_1 \ell^2 \nabla\alpha \cdot \nabla\alpha d\Omega \qquad [3.21]$$

and redefine the total energy by

$$\mathcal{E}(u, \alpha) = \int_\Omega W(\boldsymbol{\varepsilon}(u), \alpha, \nabla\alpha) d\Omega - \int_\Omega f \cdot u d\Omega - \int_{\partial T} T \cdot u dS \qquad [3.22]$$

The evolution problem consists of finding $u \in \mathcal{C}$ and $\dot{\alpha} \geq 0$ such that

$$D\mathcal{E}(u, \alpha)(v - u, \dot{\beta} - \dot{\alpha}) \geq 0, \forall v \in \mathcal{C}, \forall \dot{\beta} \geq 0 \qquad [3.23]$$

EXAMPLE 3.3.– *We will show an example, where we have a 1D bar that only breaks in a small region. For more details, the reader is referred to [MAR 11].*

We will assume that fracture occurs in the interval $[x_0-\Delta, x_0+\Delta]$, *where* x_0 *is the center of the damage profile (assumed to be symmetric) and the value of* Δ *is unknown.*

Since the whole region is damaged, α *satisfies the damage criterion in the interval:*

$$\frac{1}{2}E'(\alpha)\varepsilon^2 + w'(\alpha) - w_1\ell^2\alpha'' = 0 \qquad [3.24]$$

If we write $S(\alpha) = 1/E(\alpha)$, *then (omitting* α*)* $S' = -1/E^2E'$. *Thus,* $E' = -E^2S'$ *and*

$$-\frac{1}{2}S'(\alpha)\sigma^2 + w'(\alpha) - w_1\ell^2\alpha'' = 0 \qquad [3.25]$$

We multiply this expression by α' *and integrate to obtain*

$$-\frac{1}{2}S(\alpha)\sigma^2 + w(\alpha) - w_1\ell^2\frac{\alpha'^2}{2} = \textit{ constant } = -\frac{\sigma^2}{2E(0)} \qquad [3.26]$$

We define the function H *by*

$$H(\alpha) := \sigma^2\Big(\frac{1}{w_1E(0)} - \frac{S(\alpha)}{w_1}\Big) + \frac{2w(\alpha)}{w_1} \qquad [3.27]$$

Then

$$\ell^2\alpha'^2 = H(\alpha) \qquad [3.28]$$

The maximum value α_{max} *of* α *is such that* $H(\alpha_{max}) = 0$. *For a given* α_{max}, *we can find the value of stress* σ *on the bar:*

$$\sigma = \sqrt{\frac{2w(\alpha_{max})}{S(\alpha_{max}) - 1/E(0)}} \qquad [3.29]$$

We can then find the damage profile using the relation

$$x(\alpha) = x_0\int_{\alpha}^{\alpha_{max}} \frac{\ell}{H(\beta)}d\beta \qquad [3.30]$$

The different damage profiles are shown in Figure 3.2.

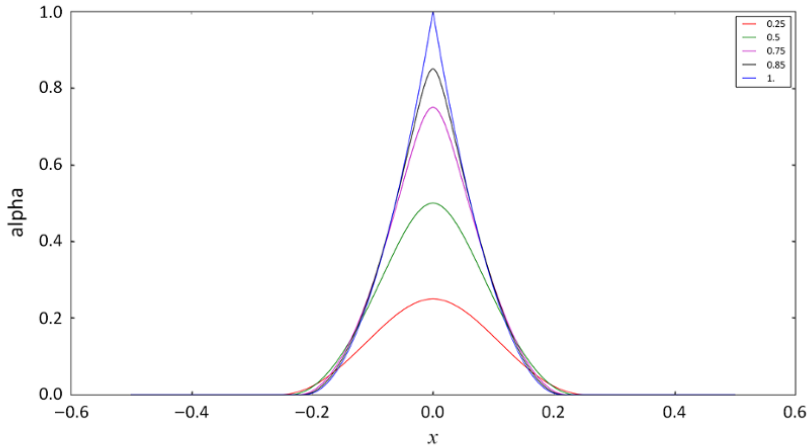

Figure 3.2. *Damage profiles for w_1=1 and $E(\alpha) = (1-\alpha)^2$ depending on the maximal value of α, given by equation [3.30]. For a color version of the figure, see www.iste.co.uk/lambert/dynamic.zip*

Consider now that $E(\alpha)$=$(1-\alpha)^2$ and $w(\alpha)$=$w_1\alpha$. When the material is completely broken, we have σ=0 and $H(\alpha) = 2\alpha$. We then solve $\ell^2\alpha'^2 = 2\alpha$, i.e.

$$\frac{\ell d\alpha}{\sqrt{\alpha}} = \sqrt{2}dx, \qquad [3.31]$$

which, considering that $\alpha(x_0) = 1$, gives

$$\sqrt{2}\ell(1-\sqrt{\alpha}) = x - x_0 \qquad [3.32]$$

We thus find for $x \in (x_0 - \Delta, x_0 + \Delta)$

$$\alpha(x) = \left(1 - \frac{|x-x_0|}{\ell\sqrt{2}}\right)^2 \quad \textit{and} \quad \Delta = \sqrt{2}\ell \qquad [3.33]$$

The dissipated energy can now be calculated as

$$\mathcal{D} = \int_{x_0-\Delta}^{x_0+\Delta} \Big(w_1(\alpha(x)) + \frac{1}{2} w_1 \ell^2 \alpha'^2 \Big) dx$$
$$= 2 \int_0^{\ell\sqrt{2}} \Big(2w_1 \big(1 - \frac{y}{\ell\sqrt{2}}\big)^2 \Big) dy \qquad [3.34]$$

Thus

$$\mathcal{D} = \frac{4\sqrt{2}}{3} w_1 \ell \qquad [3.35]$$

3.2.1.3. *Numerical implementation of damage*

In the previous subsection, we described the model for brittle damage evolution. For a given set of boundary conditions, we want to minimize the total energy $\mathcal{E}(u, \alpha)$ with respect to u and α, with the constraint $\dot{\alpha} \geq 0$ (irreversibility condition).

We will consider a problem discrete in time, i.e. we divide the time in instants $t_0, t_1, t_2, ...$, with $t_i < t_{i+1}$, and study the evolution of the system. At an instant t_i, we have a displacement u^i and damage u^i.

In the discrete problem, u^i and α^i must minimize the total energy $\mathcal{E}$ of the system. We recall that $\alpha^i(x) \in [0, 1]$, for all $x \in \Omega$, and if the damage at an iteration i is α^i, then the irreversibility condition is equivalent to $\alpha^{i+1}(x) \geq \alpha^i(x)$, for all $x \in \Omega$.

We want to minimize $\mathcal{E}$ for u and α at the same time. The functional $\mathcal{E}$ may not be convex for the pair (u, α). It is, however, convex for each variable. Hence, we use an alternate minimization [MAR 00] procedure, i.e. we alternately minimize $\mathcal{E}$ in u for a fixed α and then, for a fixed u, minimize $\mathcal{E}$ in α, until we obtain convergence for the pair (u, α).

This method only guarantees that the solution is a stationary point or, in some cases, a local minimum of $\mathcal{E}$. It does not ensure the stability of the solution.

To calculate (u^i, α^i), we have to know the values of u^{i-1} and α^{i-1}. At each time iteration i, we have to:

1) Update boundary conditions.

2) Set $(u^{(i,0)}, \alpha^{(i,0)}) := (u^{i-1}, \alpha^{i-1})$.

3) Iteration $j \geq 1$:

3.1) Solve

$$u^{(i,j)} := \text{argmin}_u \mathcal{E}(u, \alpha^{(i,j-1)});$$

3.2) Solve

$$\alpha^{(i,j)} := \text{argmin}_{\alpha^{i-1} \leq \alpha \leq 1} \mathcal{E}(u^{(i,j)}, \alpha);$$

3.3) Stop when $\|\alpha^{(i,j)} - \alpha^{(i,j-1)}\|$ is sufficiently small.

4) We then set $(u^i, \alpha^i) := (u^{(i,j)}, \alpha^{(i,j)})$.

Note: at each minimization, u and α must respect boundary conditions.

One important feature of this alternate minimization algorithm is that the energy decreases at each iteration:

$$\mathcal{E}(u^{(i,j)}, \alpha^{(i,j)}) \leq \mathcal{E}(u^{(i,j-1)}, \alpha^{(i,j-1)}) \qquad [3.36]$$

There is no guarantee that the algorithm will converge in the general case and the convergence criterion choice may influence the results. There are different aspects we can consider when choosing the convergence criterion:

– the convergence criterion depends on the variables or the total energy;

– relative tolerance or absolute tolerance;

– norm used: we consider only the difference between elements, the sum all over the structure and whether the gradient of the variables is important.

3.2.2. *Damage coupled with plasticity*

The family of models we have developed so far cannot account for residual strains. In this section, we will extend the damage models described in section 3.2.1.2 to ductile materials.

The main idea is to define the total energy of the body in terms of the displacement field, damage, plastic strain and cumulated plastic strain and study the temporal evolution of the system through the minimization with respect to each variable.

Section 3.2.1.2 was a review of the variational approach used to describe damage evolution for a quasi-static loading. We start this section by doing the same for perfect plasticity. We then present and discuss an energy expression that can couple these two phenomena and show one example of material behavior.

3.2.2.1. *Perfect plasticity model*

We will denote the plastic strain by $\boldsymbol{\varepsilon}^p$. The stress is now given by

$$\boldsymbol{\sigma} = \mathbf{E}{:}(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p) \qquad [3.37]$$

In the general case, $\boldsymbol{\sigma}$ is admissible if it satisfies $f(\boldsymbol{\sigma}) \leq 0$, where the function f depends on the criterion used. The evolution law is given by the relation

$$\|\dot{\boldsymbol{\varepsilon}^p}\| \cdot f(\boldsymbol{\sigma}) = 0 \qquad [3.38]$$

In the case of the von-Mises yield criterion, we have in 1D

$$f(\sigma) = |\sigma| - \sigma_Y \qquad [3.39]$$

and in 3D

$$f(\boldsymbol{\sigma}) = \sqrt{\frac{3}{2}\boldsymbol{s}{:}\boldsymbol{s}} - \sigma_Y, \qquad [3.40]$$

where $\boldsymbol{s} := \boldsymbol{\sigma} - \frac{\mathrm{Tr}\sigma}{3}\mathbf{I}$ is the deviatoric stress and σ_Y is the yield stress and is a material constant.

We define the cumulated plastic strain from zero to the instant t as

$$\bar{p}_t = \bar{p}_0 + \int_0^t \|\dot{\boldsymbol{\varepsilon}^p}\| d\tau \qquad [3.41]$$

and the energy density for an elasto-plastic material as

$$W^{1D}(\varepsilon, \varepsilon^p, \bar{p}) = \frac{1}{2}E(\varepsilon - \varepsilon^p)^2 + \sigma_Y \bar{p} \qquad [3.42]$$

in 1D and as

$$W^{3D}(\boldsymbol{\varepsilon}, \boldsymbol{\varepsilon}^p, \bar{p}) = \frac{1}{2}(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p):\mathbf{E}:(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p) + \sqrt{\frac{2}{3}}\sigma_Y \bar{p} \qquad [3.43]$$

in 3D.

We consider a temporal discretization $t_0, t_1, ...$, and we can calculate the cumulated plasticity as

$$\bar{p}_i = \bar{p}_{i-1} + \|(\boldsymbol{\varepsilon}^p)^i - (\boldsymbol{\varepsilon}^p)^{i-1}\| \qquad [3.44]$$

3.2.2.2. *Plasticity evolution in 1D*

We define the function as

$$f(\varepsilon, p) = \frac{1}{2}Ep^2 - E\varepsilon p + \sigma_Y |p - (\varepsilon^p)^{i-1}| \qquad [3.45]$$

It is clear that the minimization of f in p is equivalent to the minimization of $W^{1D}(\varepsilon(u), \varepsilon^p)$ in ε^p.

The function f is strictly convex in p and is differentiable everywhere except in $p = (\varepsilon^p)^{i-1}$. As a result, f has one unique minimum.

We use two auxiliary results:

PROPOSITION 3.1.– *For a given ε, set $\sigma^* = E(\varepsilon - (\varepsilon^p)^{i-1})$. The value p that minimizes $f(\varepsilon, p)$ can be characterized by:*

1) if $|\sigma^*| \leq \sigma_Y$, then the minimum is attained in $(\varepsilon^p)^{i-1}$;

2) if $|\sigma^*| > \sigma_Y$, then the minimum is attained at a point such that $\frac{\partial f}{\partial p}(\varepsilon, p) = 0$.

PROOF.– We write $p = (\varepsilon^p)^{i-1} + e$. Then

$$f(\varepsilon, p) = f(\varepsilon, (\varepsilon^p)^{i-1}) + \frac{1}{2}Ee^2 - \sigma^* e + \sigma_Y |e| \qquad [3.46]$$

1) If $|\sigma^*| \leq \sigma_Y$, then $\sigma^* e \leq \sigma_Y |e|$ and $f(\varepsilon, p) \geq f(\varepsilon, (\varepsilon^p)^{i-1}) + \frac{1}{2}Ee^2$. Hence, the minimum is attained when $e = 0$, i.e. when $p = (\varepsilon^p)^{i-1}$.

2) If $|\sigma^*| > \sigma_Y$, then we put $e = h\sigma^*/|\sigma^*|$, with $h > 0$. Then

$$f(p) = f(\varepsilon, (\varepsilon^p)^{i-1}) + \frac{1}{2}Eh^2 - \sigma^* h + \sigma_Y h \qquad [3.47]$$

If h is small enough, then $f(\varepsilon, p) < f(\varepsilon, (\varepsilon^p)^{i-1})$. Since f is regular everywhere except at $e = 0$, we must have $\frac{\partial f}{\partial p}(\varepsilon, p) = 0$.

PROPOSITION 3.2.– *In the evolution problem, we set $\sigma^* = E(\varepsilon^i - (\varepsilon^p)^{i-1})$. The minimization of W in ε^p is equivalent to:*

1) if $|\sigma^*| \leq \sigma_Y$:

$$(\varepsilon^p)^i = (\varepsilon^p)^{i-1} \qquad [3.48]$$

2) if $|\sigma^*| > \sigma_Y$:

$$(\varepsilon^p)^i = (\varepsilon^p)^{i-1} + \left(1 - \frac{\sigma_Y}{|\sigma^*|}\right)\left(\varepsilon^i - (\varepsilon^p)^{i-1}\right) \qquad [3.49]$$

and

$$\left|E(\varepsilon^i - (\varepsilon^p)^i)\right| = \sigma_Y \qquad [3.50]$$

PROOF.– We have already proved (1) in proposition 3.1.

To prove (2), again by proposition 3.1, we have to find p such that $\frac{\partial f}{\partial p}(\varepsilon^i, p) = 0$.

We note that for $e \neq 0$ and $|\delta e| < |e|$, we have

$$|e + \delta e| = |e| + \delta e \frac{e}{|e|} \qquad [3.51]$$

Then

$$\frac{\partial f}{\partial p}(\varepsilon^i, p) = Ee - \sigma^* + \sigma_Y \frac{e}{|e|} = 0 \qquad [3.52]$$

Hence

$$E(p - (\varepsilon^p)^{i-1}) - E(\varepsilon^i - (\varepsilon^p)^{i-1}) + \sigma_Y \frac{e}{|e|} = 0 \qquad [3.53]$$

Rearranging the terms

$$E(\varepsilon^{(i,j)} - p) = \sigma_Y \frac{e}{|e|} \tag{3.54}$$

If we write $\sigma = E(\varepsilon^i - p)$, then, by taking the absolute values, we obtain $|\sigma| = |\sigma_Y|$.

We can write

$$e = \frac{\sigma^* - \sigma}{E} \tag{3.55}$$

Since we are working on the case $|\sigma| = \sigma_Y < |\sigma^*|$, we have

$$\frac{e}{|e|} = \frac{\sigma^* - \sigma}{|\sigma^* - \sigma|} = \frac{\sigma^*}{|\sigma^*|} \tag{3.56}$$

Finally, by [3.53]

$$e = \frac{1}{E}\Big(\sigma^* - \sigma_Y \frac{e}{|e|}\Big) = \frac{1}{E}\Big(\sigma^* - \sigma_Y \frac{\sigma^*}{|\sigma^*|}\Big) \tag{3.57}$$

and

$$(\varepsilon^p)^i := p = (\varepsilon^p)^{i-1} + e = (\varepsilon^p)^{i-1} + \Big(1 - \sigma_Y \frac{\sigma_Y}{|\sigma^*|}\Big)(\varepsilon^i - (\varepsilon^p)^{i-1}) \tag{3.58}$$

This is an elastic prediction-plastic correction procedure: we calculate the current strain and stress based on the previous time instant assuming that the material is elastic (elastic prediction). If the stress is inside the elastic domain, i.e. $|\sigma^*| < \sigma_Y$, then we keep it and the plasticity does not change. On the other hand, if the stress is not in the elastic domain, then we update the plastic strain (plastic correction).

This behavior is shown in Figure 3.3. We can see the normalized strain $\bar{\varepsilon}{:=}\varepsilon E_0/\sigma_Y$, normalized stress and normalized plastic strain.

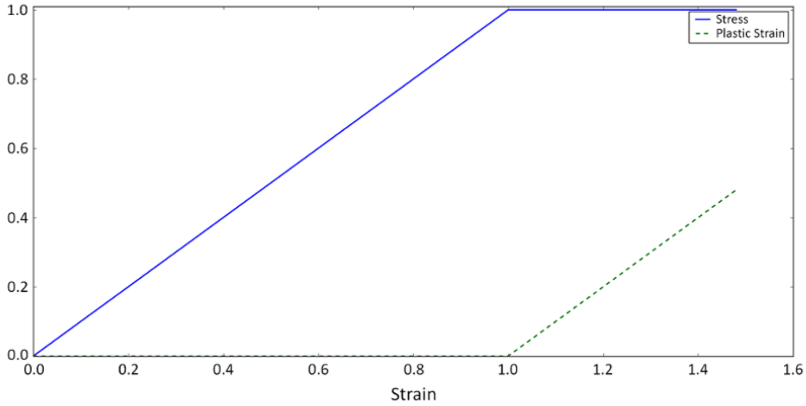

Figure 3.3. *Damage (dashed green line) and normalized stress (blue line). For a color version of the figure, see www.iste.co.uk/lambert/dynamic.zip*

3.2.2.3. *Plasticity evolution in 3D*

In this section, we will write the same results as in the previous section, but now to a problem in 3D.

We consider the usual assumption of plastic incompressibility

$$\mathrm{tr}(\boldsymbol{\varepsilon}^p) = 0 \qquad [3.59]$$

and

$$\sqrt{\frac{3}{2}\boldsymbol{s}:\boldsymbol{s}} \leq \sigma_Y, \qquad [3.60]$$

where the deviatoric stress tensor $\boldsymbol{s}$ is given by

$$\boldsymbol{s} = \boldsymbol{\sigma} - \frac{\mathrm{tr}(\boldsymbol{\sigma})}{3}\mathbf{I} \qquad [3.61]$$

The total energy density is written as

$$W(\boldsymbol{\varepsilon}, \boldsymbol{\varepsilon}^p) = \frac{1}{2}(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p):\mathbf{E}:(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p) + \sqrt{\frac{2}{3}}\sigma_Y \|\boldsymbol{\varepsilon}^p - (\boldsymbol{\varepsilon}^p)^{i-1}\|, \qquad [3.62]$$

where

$$\|e\| = \sqrt{\boldsymbol{e}:\boldsymbol{e}} \qquad [3.63]$$

We now define $\boldsymbol{e}$ as the deviatoric part of $\boldsymbol{\varepsilon}$, and since $\mathrm{tr}(\boldsymbol{\varepsilon}^p) = 0$, the minimization of W in $\boldsymbol{\varepsilon}^p$ is equivalent to

$$\min_{\{\boldsymbol{p}\,:\,\mathrm{tr}(\boldsymbol{p})=0\}} f(\boldsymbol{p}), \text{ for every point in } \Omega \qquad [3.64]$$

where

$$f(\boldsymbol{\varepsilon}, \boldsymbol{p}) := \mu \boldsymbol{p}:\boldsymbol{p} - 2\mu \boldsymbol{e}:\boldsymbol{p} + \sqrt{\frac{2}{3}}\sigma_Y \|\boldsymbol{p} - (\boldsymbol{\varepsilon}^p)^{i-1}\| \qquad [3.65]$$

(Lamé's coefficients are denoted by λ and μ.)

We set $\boldsymbol{\sigma}^* = \mathbf{E}:(\boldsymbol{\varepsilon}^i - (\boldsymbol{\varepsilon}^p)^{i-1})$, and its deviatoric part is given by $\boldsymbol{s}^* = 2\mu(\boldsymbol{e} - (\boldsymbol{\varepsilon}^p)^{i-1})$

The following propositions are the 3D equivalents of the auxiliary results in section 3.2.2.3:

PROPOSITION 3.3.– *The value p that minimizes $f(\varepsilon, p)$ can be characterized by:*

1) if $\|\boldsymbol{s}^*\| \leq \sqrt{\frac{2}{3}}\sigma_Y$, then the minimum is attained in $(\varepsilon^p)^{i-1}$;

2) if $\|\boldsymbol{s}^*\| > \sqrt{\frac{2}{3}}\sigma_Y$, then the minimum is attained at a point such that $\frac{\partial f}{\partial \boldsymbol{p}}(\boldsymbol{\varepsilon}, \boldsymbol{p}) = 0$.

PROOF.– We write $\boldsymbol{p} = (\boldsymbol{\varepsilon}^p)^{i-1} + \boldsymbol{\delta}$. Then

$$\begin{aligned} f(\boldsymbol{\varepsilon}, \boldsymbol{p}) := \mu \boldsymbol{p}:\boldsymbol{p} - 2\mu \boldsymbol{e}:\boldsymbol{p} + \sqrt{\frac{2}{3}}\sigma_Y \|p - (\boldsymbol{\varepsilon}^p)^{i-1}\| \\ + f((\boldsymbol{\varepsilon}^p)^{i-1}) - \mu(\boldsymbol{\varepsilon}^p)^{i-1}:(\boldsymbol{\varepsilon}^p)^{i-1} + 2\mu \boldsymbol{e}:(\boldsymbol{\varepsilon}^p)^{i-1} \end{aligned}$$

$$= f((\boldsymbol{\varepsilon}^p)^{i-1}) + \mu\boldsymbol{\delta}{:}\boldsymbol{\delta} + 2\mu\boldsymbol{p}{:}(\boldsymbol{\varepsilon}^p)^{i-1}$$

$$- 2\mu(\boldsymbol{\varepsilon}^p)^{i-1}{:}(\boldsymbol{\varepsilon}^p)^{i-1} - 2\mu\boldsymbol{e}{:}\boldsymbol{\delta} + \sqrt{\frac{2}{3}}\sigma_Y\|\boldsymbol{\delta}\| \quad [3.66]$$

$$= f((\boldsymbol{\varepsilon}^p)^{i-1}) + \mu\boldsymbol{\delta}{:}\boldsymbol{\delta} + 2\mu\boldsymbol{\delta}{:}(\boldsymbol{\varepsilon}^p)^{i-1} - 2\mu\boldsymbol{e}{:}\boldsymbol{\delta} + \sqrt{\frac{2}{3}}\sigma_Y\|\boldsymbol{\delta}\|$$

$$= f((\boldsymbol{\varepsilon}^p)^{i-1}) + \mu\boldsymbol{\delta}{:}\boldsymbol{\delta} + \sqrt{\frac{2}{3}}\sigma_Y\|\boldsymbol{\delta}\| - \boldsymbol{s}^*{:}\boldsymbol{\delta}$$

1) If $\|\boldsymbol{s}^*\| \leq \sqrt{\frac{2}{3}}\sigma_Y$, then $f(\boldsymbol{\varepsilon},\boldsymbol{p}) \geq f(\boldsymbol{\varepsilon},(\boldsymbol{\varepsilon}^p)^{i-1}) + \mu\boldsymbol{\delta}{:}\boldsymbol{\delta}$. Hence, the minimum is attained when $\boldsymbol{\delta} = \mathbf{0}$.

2) If $\|\boldsymbol{s}^*\| > \sqrt{\frac{2}{3}}\sigma_Y$, then we put $\boldsymbol{\delta} = h\boldsymbol{s}^*/\|\boldsymbol{s}^*\|$, with $h > 0$. Then

$$f(\boldsymbol{p}) = f((\boldsymbol{\varepsilon}^p)^{i-1}) + \sqrt{\frac{2}{3}}\sigma_Y h - \|\boldsymbol{s}^*\|h + \mu h^2 \quad [3.67]$$

If h is small enough, then $f(\boldsymbol{p}) < f((\boldsymbol{\varepsilon}^p)^{i-1})$. Since f is regular everywhere, except $\boldsymbol{\delta} = \mathbf{0}$, we must have $\frac{\partial f}{\partial \boldsymbol{p}} f(\boldsymbol{p}) = \mathbf{0}$. □

PROPOSITION 3.4.– *The minimization of W in $\boldsymbol{\varepsilon}^p$ is equivalent to:*

1) if $\|\boldsymbol{s}^*\| \leq \sqrt{\frac{2}{3}}\sigma_Y$:

$$(\boldsymbol{\varepsilon}^p)^i = (\boldsymbol{\varepsilon}^p)^{i-1} \quad [3.68]$$

2) if $\|\boldsymbol{s}^*\| > \sqrt{\frac{2}{3}}\sigma_Y$:

$$(\boldsymbol{\varepsilon}^p)^i = (\boldsymbol{\varepsilon}^p)^{i-1} + \left(1 - \frac{\sqrt{\frac{2}{3}}\sigma_Y}{|\boldsymbol{s}^*|}\right)\left(\boldsymbol{e}^i - (\boldsymbol{\varepsilon}^p)^{i-1}\right) \quad [3.69]$$

PROOF.– The proof of (1) follows directly from the last proposition.

To prove (2), we have to find $\boldsymbol{p}$ such that $\frac{\partial f}{\partial \boldsymbol{p}}(\boldsymbol{\varepsilon},\boldsymbol{p}) = \mathbf{0}$.

We derive f and apply it to a tensor $\boldsymbol{\delta}$:

$$\frac{\partial f}{\partial \boldsymbol{p}}(\boldsymbol{\varepsilon},\boldsymbol{p}){:}\boldsymbol{\delta} = 2\mu(\boldsymbol{p} - \boldsymbol{e}^i){:}\boldsymbol{\delta} + \sqrt{\frac{2}{3}}\sigma_Y \frac{\boldsymbol{p} - (\boldsymbol{\varepsilon}^p)^{i-1}}{\|\boldsymbol{p} - (\boldsymbol{\varepsilon}^p)^{i-1}\|}{:}\boldsymbol{\delta} = \mathbf{0} \quad [3.70]$$

If

$$\boldsymbol{\sigma}^i = \mathbf{E}:(\boldsymbol{\varepsilon}^i - \boldsymbol{p}) \qquad [3.71]$$

and $\boldsymbol{s}^i$ is its deviatoric part, then we must have

$$\boldsymbol{s}^i = \sqrt{\frac{2}{3}}\sigma_Y \frac{\boldsymbol{p} - (\boldsymbol{\varepsilon}^p)^{i-1}}{\|\boldsymbol{p} - (\boldsymbol{\varepsilon}^p)^{i-1}\|} \qquad [3.72]$$

It is clear that

$$\|\boldsymbol{s}^i\| = \sqrt{\frac{2}{3}}\sigma_Y \qquad [3.73]$$

We note that

$$\boldsymbol{s}^i = \boldsymbol{s}^* + 2\mu((\boldsymbol{\varepsilon}^p)^{i-1} - \boldsymbol{p}) \qquad [3.74]$$

and, by equation [3.72], $\boldsymbol{s}^i$ and $\boldsymbol{s}^*$ have the same direction.

Since we know $\boldsymbol{s}^*$, we obtain

$$\boldsymbol{s}^i = \sqrt{\frac{2}{3}}\sigma_Y \frac{\boldsymbol{s}^*}{\|\boldsymbol{s}^*\|} \qquad [3.75]$$

Finally, applying this to [3.72] and [3.74]

$$\begin{aligned}\boldsymbol{p} - (\boldsymbol{\varepsilon}^p)^{i-1} &= \boldsymbol{s}^i \frac{\|\boldsymbol{p} - \boldsymbol{\varepsilon}^{i-1}\|}{\sqrt{\frac{2}{3}}\sigma_Y} = \boldsymbol{s}^i \frac{\|\boldsymbol{s}^i - \boldsymbol{s}^*\|}{2\mu\sqrt{\frac{2}{3}}\sigma_Y} = \boldsymbol{s}^* \frac{\|\boldsymbol{s}^i - \boldsymbol{s}^*\|}{2\mu\|\boldsymbol{s}^*\|} \\ &= \left(1 - \frac{\sqrt{\frac{2}{3}}\sigma_Y}{\|\boldsymbol{s}^*\|}\right)\frac{\boldsymbol{s}^*}{2\mu} = \left(1 - \frac{\sqrt{\frac{2}{3}}\sigma_Y}{\|\boldsymbol{s}^*\|}\right)(\boldsymbol{e}^i - (\boldsymbol{\varepsilon}^p)^{i-1})\end{aligned} \qquad [3.76]$$

We conclude by taking $(\boldsymbol{\varepsilon}^p)^i = \boldsymbol{p}$.

3.2.2.4. *Damage–plasticity coupling*

In this section, in order to construct a family of models that account for plasticity and damage, instead of proposing the evolution laws for each variable, we work directly with a suitable form of energy and, by minimizing this energy, we deduce the constitutive relations. For simplicity, we remove volume forces from our calculations

In section 3.2.1.2, a total energy for brittle damage was postulated:

$$\mathcal{E}_{brittle}(u,\alpha) = \int_\Omega \Big(\psi(\alpha, \boldsymbol{\varepsilon}(u)) + w(\alpha) + \frac{1}{2} w_1 \ell^2 \nabla\alpha \cdot \nabla\alpha\Big) d\Omega \quad [3.77]$$

We recall that the evolution of the system for quasi-static loading can be obtained by minimizing this energy with respect to u and α. A perturbation in the direction u gives us the static equilibrium, and a perturbation in α gives us the damage criterion.

In section 3.2.2.1, we showed that the evolution of the plasticity minimizes the energy

$$\mathcal{E}^{1D}_{plast}(\varepsilon,\varepsilon^p) = \int_\Omega \Big(\frac{1}{2}E(\varepsilon - \varepsilon^p)^2 + \sigma_Y \bar{p}\Big) d\Omega \quad [3.78]$$

in 1D and

$$\mathcal{E}^{3D}_{plast}(\boldsymbol{\varepsilon},\boldsymbol{\varepsilon}^p) = \int_\Omega \Big(\frac{1}{2}(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p):\mathbf{E}:(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p) + \sqrt{\frac{2}{3}}\sigma_Y \bar{p}\Big) d\Omega \quad [3.79]$$

in 3D. By examining perturbations in $\boldsymbol{\varepsilon}$ and $\boldsymbol{\varepsilon}^p$, we obtain the static equilibrium and the plasticity criterion, respectively.

As we can see, the problems for damage and plasticity are similar in the sense that the quasi-static evolution in both cases is found after minimizing the total energy. For the coupled problem, we will use an energy form that is, in a way, a combination of the damage energy and the plastic energy. For that, we will assume that the yield stress now depends on the damage, i.e. $\sigma_Y{=}\sigma_Y(\alpha)$.

We define the following 1D and 3D energies for the damage–plasticity (DP) coupling:

$$\mathcal{E}^{1D}_{DP}(\varepsilon,\varepsilon^p,\bar{p},\alpha) = \int_\Omega \Big(\frac{1}{2}E(\alpha)(\varepsilon - \varepsilon^p)^2 + \sigma_Y(\alpha)\bar{p} + w(\alpha) + \frac{1}{2}w_1\ell^2\alpha'^2\Big) d\Omega \quad [3.80]$$

and

$$\mathcal{E}_{DP}^{3D}(\boldsymbol{\varepsilon},\boldsymbol{\varepsilon}^p,\bar{p},\alpha) = \int_\Omega \Big(\frac{1}{2}(\boldsymbol{\varepsilon}-\boldsymbol{\varepsilon}^p){:}\mathbf{E}(\alpha){:}(\boldsymbol{\varepsilon}-\boldsymbol{\varepsilon}^p) + \sqrt{\frac{2}{3}}\sigma_Y(\alpha)\bar{p} + w(\alpha) + \frac{1}{2}w_1\ell^2|\nabla\alpha|^2\Big)d\Omega \qquad [3.81]$$

To obtain the evolution criterion, we minimize the total energy with respect to three variables (u, $\boldsymbol{\varepsilon}^p$ and α):

– The minimization of the displacement gives the static-equilibrium:

$$\mathrm{div}\boldsymbol{\sigma} = 0, \quad \text{where} \quad \boldsymbol{\sigma} = \mathbf{E}(\alpha){:}(\boldsymbol{\varepsilon}-\boldsymbol{\varepsilon}^p) \qquad [3.82]$$

– The minimization of the plastic strain gives

$$\sqrt{\frac{3}{2}\boldsymbol{s}{:}\boldsymbol{s}} \leq \sigma_Y(\alpha) \quad \text{and} \quad \|\dot{\boldsymbol{\varepsilon}}^p\| \cdot \Big(\sqrt{\frac{3}{2}\boldsymbol{s}{:}\boldsymbol{s}} - \sigma_Y(\alpha)\Big) = 0 \qquad [3.83]$$

– The minimization of α gives the new damage criterion (after taking the derivative with respect to α and integrating by parts). In 1D:

$$\frac{1}{2}E(\alpha')(\varepsilon-\varepsilon^p)^2 + \sigma_Y'(\alpha)\bar{p} + w(\alpha') - w_1\ell^2\alpha'' \geq 0 \qquad [3.84]$$

In 3D:

$$\frac{1}{2}(\boldsymbol{\varepsilon}-\boldsymbol{\varepsilon}^p){:}\mathbf{E}'(\alpha){:}(\boldsymbol{\varepsilon}-\boldsymbol{\varepsilon}^p) + \sqrt{\frac{2}{3}}\sigma_Y'(\alpha)\bar{p} + w'(\alpha) - w_1\ell^2\Delta\alpha \geq 0 \qquad [3.85]$$

We also have $\dot{\alpha} = 0$ when we have a strict inequality.

EXAMPLE 3.4.– *Consider a bar given by $\Omega{=}[0, L]$ under traction, where the displacement at the extremities is controlled. We want to calculate the evolution of damage and plastic strain for the homogeneous case. We consider $\sigma_Y^0 < \sqrt{w_1 E_0}$. We take the functions*

$$E(\alpha) = E_0(1-\alpha)^2, \qquad w(\alpha) = w_1\alpha, \qquad \sigma_Y(\alpha) = \sigma_Y^0(1-\alpha)^2 \qquad [3.86]$$

Since we are assuming uniformity in space, we have to calculate the scalars σ, ε, ε^p and α.

We have three different stages:

– Elastic phase: it is easy to see that while $\varepsilon < \sqrt{\sigma_Y^0/E_0}$*,* $\sigma < \sigma_Y(\alpha){=}\sigma_Y^0$ *and there is no change in the plastic strain. Since there is no plastic strain, the damage criterion is the same for brittle materials and we see that the bar does not suffer any damage.*

– Plastic phase: if $\varepsilon > \sqrt{\sigma_Y^0/E_0}$*, then plastic strain evolves. In a pure traction test, the plastic strain and the cumulated strain are the same and we must have* $E_0(\varepsilon{-}\varepsilon^p){=}\sigma_Y^0$*. Thus,* $\varepsilon^p{=}\varepsilon{-}\sigma_Y^0/E_0$*.*

The damage criterion becomes

$$-(1-\alpha)\frac{(\sigma_Y^0)^2}{E_0} - 2(1-\alpha)\sigma_Y^0\bar{p} + w_1 \geq 0 \qquad [3.87]$$

It is easy to see that for $\alpha{=}0$*, we have a strict inequality while* $\varepsilon^p{<}\frac{w_1}{2\sigma_Y^0} - \frac{\sigma_Y^0}{2E_0}$*.*

– Damage-plastic phase: the plasticity continues to evolve and the plastic evolution criterion gives $\varepsilon^p{=}\varepsilon{-}\sigma_Y^0/E_0$*.*

The damage criterion is now

$$-(1-\alpha)\frac{(\sigma_Y^0)^2}{E_0} - 2(1-\alpha)\sigma_Y^0\bar{p} + w_1 = 0 \qquad [3.88]$$

We can thus find

$$\alpha = \frac{\frac{\sigma_Y^0}{E_0} + 2\bar{p} - \frac{w_1}{\sigma_Y^0}}{\frac{\sigma_Y^0}{E_0} + 2\bar{p}} \qquad [3.89]$$

Figure 3.4 shows these three phases. We see the normalized (in function of damage threshold) stress $\bar{\sigma} = \sigma/\sigma_c$ *and strain* $\bar{\varepsilon} = \sigma/\varepsilon_c$*. We can clearly identify the three phases in the stress curve.*

3.2.3. *Dynamic gradient damage*

To formulate the evolution of the dynamic system, we will use the principle of least action, as in [LI 16a].

Suppose we have a mechanic system Ω whose displacement is u and stress is $\sigma(u)$. At each instant $t \in [t_1, t_2]$, we impose a displacement $u_D(t)$ on $\partial_u \subset$

$\partial\Omega$ and a normal stress $T(t)$ on $\partial_T \subset \partial\Omega$. We also suppose that $\partial_u \bigcap \partial_T = \emptyset$ and $\partial_u \bigcup \partial_T = \partial\Omega$. We have the following equations:

$$\begin{cases} \rho\ddot{u} = \mathrm{div}\boldsymbol{\sigma} + f \quad \text{on } \Omega \\ u = u_D(t) \quad \text{on } \partial_u \\ \boldsymbol{\sigma}{\cdot}n = T(t) \quad \text{on } \partial_T. \end{cases} \qquad [3.90]$$

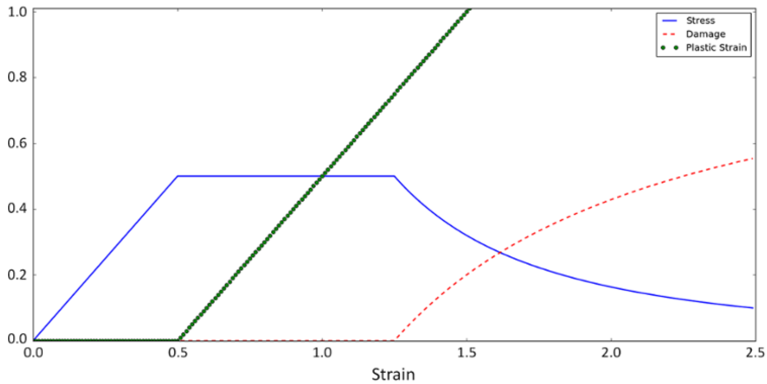

Figure 3.4. *Damage (dashed red line), normalized stress (blue line) and plastic strain (green dots). For a color version of the figure, see www.iste.co.uk/lambert/dynamic.zip*

We fix a test function w such that $w(x,t){=}0$ on ∂_u for all $t \in [t_1, t_2]$ and $w(t{=}t_1) = w(t{=}t_2) = 0$ on Ω. Then

$$\int_\Omega \rho\ddot{u}{\cdot}w d\Omega = \int_\Omega \big(\mathrm{div}\boldsymbol{\sigma}(u){\cdot}w + f{\cdot}w\big) d\Omega \qquad [3.91]$$

and Green's formula shows that

$$\int_\Omega \rho\ddot{u} \cdot w d\Omega = \underbrace{\int_{\partial u} (\boldsymbol{\sigma}{\cdot}n){\cdot}w dA}_{0} + \int_{\partial T} T{\cdot}w dA$$
$$- \int_\Omega \boldsymbol{\sigma}(u){:}\boldsymbol{\varepsilon}(w) d\Omega + \int_\Omega f{\cdot}w d\Omega \qquad [3.92]$$

We integrate this equation between instants t_1 and t_2, and after an integration by parts, we obtain

$$\Big(\int_\Omega \rho\dot{u}\cdot w d\Omega\Big)\Big|_{t_1}^{t_2} - \int_{t_1}^{t_2}\Big(\int_\Omega \rho\dot{u}\cdot\dot{w} d\Omega\Big)dt = \int_{t_1}^{t_2}\Big(\int_{\partial T} T\cdot w dA\Big)dt$$
$$-\int_{t_1}^{t_2}\Big(\int_\Omega \boldsymbol{\sigma}(u):\boldsymbol{\varepsilon}(w)d\Omega\Big)dt + \int_{t_1}^{t_2}\Big(\int_\Omega f\cdot w d\Omega\Big)dt \qquad [3.93]$$

We define the kinetic energy of the system

$$\mathcal{K}(\dot{u}) = \int_\Omega \frac{1}{2}\rho\|\dot{u}\|^2 d\Omega \qquad [3.94]$$

and the potential energy

$$\mathcal{P}(u) = \frac{1}{2}\int_\Omega \boldsymbol{\sigma}(u):\boldsymbol{\varepsilon}(u)d\Omega - \int_\Omega f\cdot u d\Omega - \int_{\partial_T} T\cdot u dA \qquad [3.95]$$

Applying the boundary conditions of w in t_1 and t_2, we have

$$\int_{t_1}^{t_2}\Big(\frac{\partial\mathcal{P}}{\partial u}w - \frac{\partial\mathcal{K}}{\partial\dot{u}}\dot{w}\Big)dt = 0, \forall w \qquad [3.96]$$

We have thus shown that the problem [3.90] implies equation [3.96]. It is easy to see that equation [3.96] can be obtained by searching for stationary points of an action functional defined by

$$\mathcal{S}(u,\dot{u}) = \int_{t_1}^{t_2} \mathcal{P}(u(t)) - \mathcal{K}(\dot{u}(t))dt \qquad [3.97]$$

This motivates us to construct a dynamic gradient model by defining a suitable form of the action functional. Instead of using a purely elastic energy $\int \boldsymbol{\sigma}:\boldsymbol{\varepsilon}$, we will use the energies defined by equations [3.80] and [3.81] with the terms containing the plastic strain and energy dissipated by the damage process. We recall that they were written as

$$\mathcal{E}_{DP}^{1D}(\varepsilon,\varepsilon^p,\bar{p},\alpha) = \int_\Omega \Big(\frac{1}{2}E(\alpha)(\varepsilon-\varepsilon^p)^2 + \sigma_Y(\alpha)\bar{p} + w(\alpha) + \frac{1}{2}w_1\ell^2\alpha'^2\Big)d\Omega$$

and

$$\mathcal{E}^{3D}_{DP}(\boldsymbol{\varepsilon}, \boldsymbol{\varepsilon}^p, \bar{p}, \alpha) = \int_\Omega \Big(\frac{1}{2}(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p) : \mathbf{E}(\alpha) : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p) + \sqrt{\frac{2}{3}} \sigma_Y(\alpha)\bar{p} + w(\alpha) + \frac{1}{2} w_1 \ell^2 |\nabla \alpha|^2 \Big) d\Omega$$

We take the external loads into account and define a potential energy as

$$\mathcal{P}_{DP}(u, \boldsymbol{\varepsilon}^p, \alpha) = \mathcal{E}_{DP}(\boldsymbol{\varepsilon}(u), \boldsymbol{\varepsilon}^p, \bar{p}, \alpha) - \int_\Omega f \cdot u d\Omega - \int_{\partial_T} T \cdot u dA \qquad [3.98]$$

We define the new Lagrangian by

$$\mathcal{L}_{DP}(u, \dot{u}, \boldsymbol{\varepsilon}^p, \bar{p}, \alpha, t) = \mathcal{P}_{DP}(u(t), \boldsymbol{\varepsilon}^p(t), \bar{p}, \alpha(t)) - \mathcal{K}(\dot{u}(t)) \qquad [3.99]$$

and the action by

$$\mathcal{S}_{DP}(u, \dot{u}, \boldsymbol{\varepsilon}^p, \bar{p}, \alpha) = \int_{t_1}^{t^2} \mathcal{L}_{DP}(u, \dot{u}, \boldsymbol{\varepsilon}^p, \bar{p}, \alpha, t) dt \qquad [3.100]$$

We define the admissible displacement space $\mathcal{C}$ and the admissible damage space $\mathcal{D}$ by

$$\mathcal{C} = \{u : u(t) = u_0(t) \text{ on } \partial_u\}$$
$$\mathcal{D} = \{\alpha \in [0,1] : \dot{\alpha} \geq 0 \text{ on } \Omega\} \qquad [3.101]$$

In order to preserve the irreversibility of damage and plasticity, instead of searching for stationary points, we will now only consider the unilateral minimal condition of the action, i.e. we search a displacement $u \in \mathcal{C}$, damage $\alpha \in \mathcal{D}$ and ε^p such that

$$\mathcal{S}_{DP}(u, \dot{u}, \boldsymbol{\varepsilon}^p, \bar{p}, \alpha) \leq \mathcal{S}_{DP}(w, \dot{w}, \boldsymbol{p}, \|\boldsymbol{p} - \boldsymbol{\varepsilon}^p\| + \bar{p}, \beta) \qquad [3.102]$$

for any $w \in \mathcal{C}$, $\beta \in \mathcal{D}$ and $\boldsymbol{p}$.

In particular, if we take $\beta = \alpha$ and $\boldsymbol{p} = \boldsymbol{\varepsilon}^p$, we must have

$$\frac{\partial \mathcal{S}_{DS}}{\partial u}(w - u) + \frac{\partial \mathcal{S}_{DS}}{\partial \dot{u}}(\dot{w} - \dot{u}) = 0 \qquad [3.103]$$

and, by following the previous calculations in reverse order, we find the problem given by [3.90].

We now set $w{=}u$ and $\boldsymbol{p}{=}\boldsymbol{\varepsilon}^p$ to study the damage evolution. If at an instant t the damage is α_t, then we define the admissible damage $\mathcal{D}_t$ taking α_t and the irreversibility condition into account:

$$\mathcal{D}_t = \{\beta : \dot{\beta} \geq 0 \text{ and } \beta \geq \alpha_t \text{ on } \Omega\} \qquad [3.104]$$

For every $\beta{\in}\mathcal{D}_t$

$$\frac{\partial \mathcal{S}_{DS}}{\partial \alpha}(\beta - \alpha) \geq 0 \qquad [3.105]$$

From this, it is easy to see that we obtain the same damage criterion for dynamic configurations and quasi-static loading:

$$\frac{\partial \mathcal{E}_{DP}}{\partial \alpha}(u, \boldsymbol{\varepsilon}^p, \bar{p}, \alpha){\cdot}(\beta - \alpha) \geq 0 \qquad [3.106]$$

Finally, the plastic evolution is obtained by taking $w{=}u$ and $\beta{=}\alpha$. Then, for any $\boldsymbol{p}$, we must have

$$\mathcal{S}_{DP}(u, \dot{u}, \boldsymbol{\varepsilon}^p, \bar{p}, \alpha) \leq \mathcal{S}_{DP}(u, \dot{u}, \boldsymbol{p}, \|\boldsymbol{p} - \boldsymbol{\varepsilon}^p\| + \bar{p}, \alpha),$$

which is the same criterion used in the quasi-static case, i.e. for any $\boldsymbol{p}$

$$\mathcal{E}_{DP}(u, \boldsymbol{\varepsilon}^p, \bar{p}, \alpha) \leq \mathcal{E}_{DP}(u, \boldsymbol{p}, \|\boldsymbol{p} - \boldsymbol{\varepsilon}^p\| + \bar{p}, \alpha) \qquad [3.107]$$

The whole set of equations can now be written as:

– Dynamic evolution:

$$\begin{cases} \rho\ddot{u} = \operatorname{div}\boldsymbol{\sigma} + f \quad \text{on } \Omega \\ u = u_D(t) \quad \text{on } \partial_u \\ \boldsymbol{\sigma}{\cdot}n = T(t) \quad \text{on } \partial_T \end{cases} \qquad [3.108]$$

– Damage evolution: for any $\beta{\geq}0$ admissible, we have

$$\frac{\partial \mathcal{E}_{DP}}{\partial \alpha}(u, \boldsymbol{\varepsilon}^p, \bar{p}, \alpha){\cdot}(\beta - \alpha) \geq 0 \qquad [3.109]$$

– Evolution of plastic strain: for any $\boldsymbol{p}$, we have

$$\mathcal{E}_{DP}(u, \boldsymbol{\varepsilon}^p, \bar{p}, \alpha) \leq \mathcal{E}_{DP}(u, \boldsymbol{p}, \|\boldsymbol{p} - \boldsymbol{\varepsilon}^p\| + \bar{p}, \alpha) \qquad [3.110]$$

3.3. Numerical implementation

To calculate the evolution of the system, we have to solve three sets of equations:

1) *Dynamic equation.* This hyperbolic differential equation gives the acceleration of the system, which can be integrated to obtain the velocity and displacement. It is the only equation where there is a dependence on time and stability is the main issue.

2) *Damage evolution.* This is an elliptic partial differential equation that must be solved globally (because of the gradient of damage) each time.

3) *Plastic evolution.* Since this is a local problem, it must be solved on each element independently.

The discrete time can be represented by the instants $t_0 < t_1 < t_2$..., and we use standard finite element discretization for spatial representation. The displacement and damage fields consist of polynomials of degree one ($P1$ elements), and the stress, total strain and plastic strain are constant on each element.

To solve the dynamic equation, we will use a slightly modified version of the explicit Newmark (central differences) scheme, since it is easy to implement, the total energy is conserved and every term can be obtained explicitly, resulting in fast calculations. The downside is that the stability condition forces the time Δt to be small and decreases when we refine the mesh [ALL 12].

Once we have a new displacement field u^i at the instant t_i, we search for α and ε^p that solve equations [3.109] and [3.110] simultaneously. For that, we use an alternate minimization between plastic strain and damage. We note that when $E(\cdot)$ and $\sigma_Y(\cdot)$ depend on the same way of α, only one iteration is needed.

To study the evolution of the plastic strain, we can use the algorithms described in sections 3.2.2.2 and 3.2.2.3. It is, however, important to note that σ_Y is still a constant in the damage problem, since it depends only on α and α is fixed during this step.

We consider the displacement u^{i-1}, velocity v^{i-1}, plastic strain $(\varepsilon^p)^{i-1}$ and damage α^{i-1} at the instant t_{i-1} to be known.

1) Calculate the acceleration a^i:

$$\rho a^{i-1} = \text{div}\boldsymbol{\sigma} = \text{div}\big(\mathbf{E}(\alpha^{i-1}){:}(\boldsymbol{\varepsilon}(u^{i-1}) - (\boldsymbol{\varepsilon}^p)^{i-1})\big)$$

2) Update the displacement:

$$u^i = u^{i-1} + \Delta t v^{i-1} + \frac{\Delta t^2}{2} a^{i-1}$$

3) Set $((\boldsymbol{\varepsilon}^p)^{(i,0)}, \alpha^{(i,0)}) := ((\boldsymbol{\varepsilon}^p)^{(i-1)}, \alpha^{i-1})$

4) Iteration $j \geq 1$:

4.1) Using sections 3.2.2.2 and 3.2.2.3, solve:

$$(\boldsymbol{\varepsilon}^p)^{(i,j)} := \text{argmin}_{\boldsymbol{p}} \mathcal{E}_{DP}(u^i, \boldsymbol{p}, \bar{p}_{i-1} + \|\boldsymbol{p} - (\boldsymbol{\varepsilon}^p)^{i-1}\|, \alpha^{(i,j-1)})$$

4.2) Solve:

$$\alpha^{(i,j)} := \text{argmin}_{\alpha^{i-1} \leq \alpha \leq 1} \mathcal{E}_{DP}(u^i, (\boldsymbol{\varepsilon}^p)^{(i,j)}, \bar{p}_{i-1} + \|(\boldsymbol{\varepsilon}^p)^{(i,j)} - (\boldsymbol{\varepsilon}^p)^{i-1}\|, \alpha)$$

4.3) Stop when $\|(\boldsymbol{\varepsilon}^p)^{(i,j)} - (\boldsymbol{\varepsilon}^p)^{(i,j-1)}\|$ and $\|\alpha^{(i,j)} - \alpha^{(i,j-1)}\|$ are sufficiently small.

5) We then set $((\boldsymbol{\varepsilon}^p)^i, \alpha^i) := ((\boldsymbol{\varepsilon}^p)^{(i,j)}, \alpha^{(i,j)})$ and $\bar{p}_i {=} \bar{p} + \|(\boldsymbol{\varepsilon}^p)^{(i,j)} - (\boldsymbol{\varepsilon}^p)^{i-1}\|$.

6) Calculate the acceleration a^{i+1} using:

$$\rho a^{i+1} = \text{div}\boldsymbol{\sigma} = \text{div}\big(\mathbf{E}(\alpha^{i+1}){:}(\boldsymbol{\varepsilon}(u^{i+1}) - (\boldsymbol{\varepsilon}^p)^{i+1})\big)$$

7) Update the velocity:

$$v^{i+1} = v^i + \frac{\Delta t}{2}(a^i + a^{i+1})$$

3.4. Applications

In this first set of examples, we want to show the influence of plasticity and dynamics.

3.4.1. *1D fracture*

We first study a 1D bar under traction. We have already studied this behavior for a quasi-static loading and we were able to break the bar, where damage is localized in one region. The bar was then split into two bars, and each bar had zero stress.

In dynamics, however, we have to take the inertial effects into consideration. When the bar breaks, the bar is not in static equilibrium and there are waves that propagate in the bar and cause the bar to continue breaking, even after the first crack.

Another interesting phenomenon is the importance of the waves and vibrations. When a wave reaches one of the extremities, the wave is reflected and a compression stress becomes a traction stress and vice versa. When two waves interpose each other, the resulting wave can have a greater amplitude than the initial waves.

We show here a simple example that illustrates this behavior. Damage evolution will only be considered when the stress is positive and the material is assumed to be brittle.

We will study the shock between two bars. The first bar, on the left, has size L and initial speed $v_0 > 0$. This bar hits a second bar, of size $2L$, which is initially at rest. The two bars are made of the same material and have the same thickness.

A shock wave then propagates between the bars, causing a crack in the middle of the larger bar.

In our model, we consider a single bar of length $3L$. The interval $[0, L]$ has initial speed v_0, and $[L, 3L]$ has zero initial speed. The stress is supposed to be zero at the extremities. The damage profile can be seen in Figure 3.5.

3.4.2. *Material behavior*

In this section, we present a small and far from extensive list of material behaviors that we can obtain only by changing how the functions $\mathbf{E}(\cdot)$, $w(\cdot)$ and $\sigma_Y(\cdot)$ depend on α. In these simulations, we suppose that the system uniformly evolves in space.

In Figure 3.6, we have $E(\alpha)=(1-\alpha)^2$ and $w(\alpha)=\alpha$ and we do not have plastic strain. We can clearly see an elastic phase and then a phase where damage evolves. By taking into account the plastic evolution (Figure 3.7), we see that we now have three phases (elastic, plastic with no damage and plastic with damage). It is important to note that, for both models, the stress is maximal before the beginning of the damage phase and then it decreases until it reaches zero.

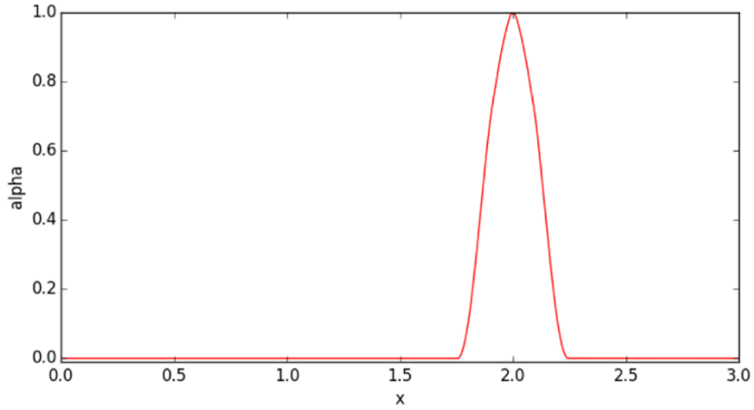

Figure 3.5. *Damage profile after failure in a 1D bar, as described in subsection 3.4.1*

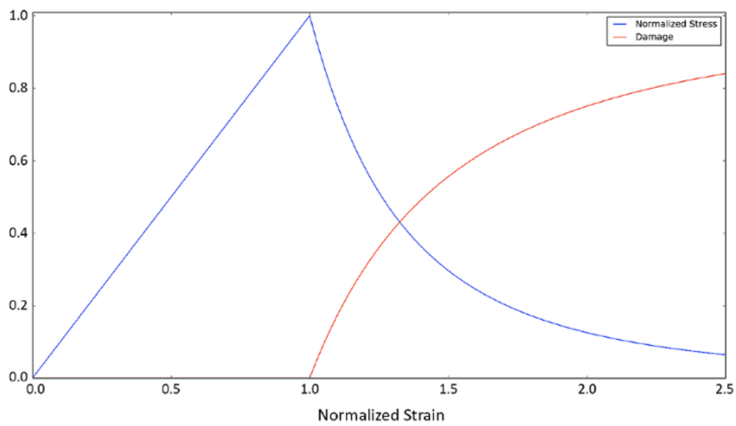

Figure 3.6. *Evolution of stress and damage for a 1D brittle material for the case $E(\alpha)=(1-\alpha)^2$ and $w(\alpha)=\alpha$; plastic strains neglected. For a color version of the figure, see www.iste.co.uk/lambert/dynamic.zip*

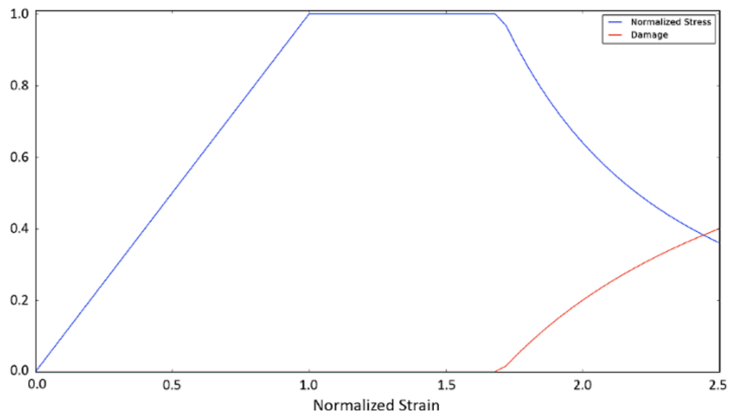

Figure 3.7. *Evolution of stress and damage for a 1D ductile material (using the von Mises criterion) for the case $E(\alpha){=}\sigma_Y(\alpha){=}(1-\alpha)^2$ and $w(\alpha){=}\alpha$. For a color version of the figure, see www.iste.co.uk/lambert/dynamic.zip*

For this next set of models, where we take $w(\alpha){=}\alpha^2$, we see that the behavior changes. In Figure 3.8, we see the evolution of brittle damage. There is no longer an elastic phase and, as strain increases, both damage and stress increase, even though the stress–strain relationship is no longer linear because of damage evolution.

Many other evolution laws could be proposed by taking, for instance, a different polynomial degree for the previous expressions or by combining them.

3.4.3. *Dimensionless parameters*

In this first example, we will study a bar made of a brittle material under traction. The objective here is to show that the brittle damage model in question depends only on two dimensionless parameters.

We consider a bar $\Omega = [0, L]$. We write $E(\alpha) = E_0 a(\alpha)$, where $a(\alpha = 0) = 1$ and $a(\alpha = 1) = 0$.

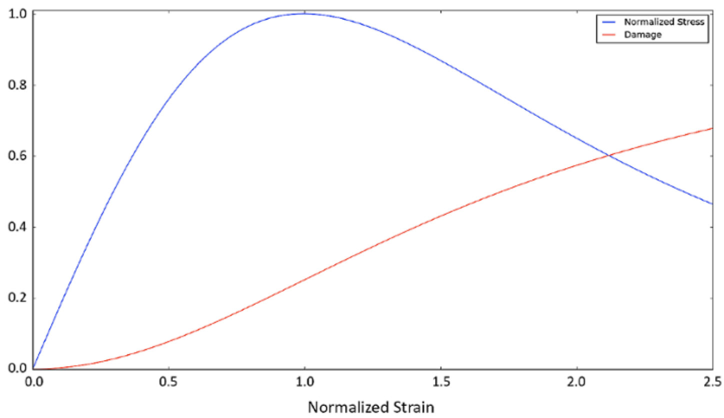

Figure 3.8. *Evolution of stress and damage for a 1D brittle material for the case $E(\alpha)=(1-\alpha)^2$ and $w(\alpha)=\alpha^2$; plastic strains neglected. For a color version of the figure, see www.iste.co.uk/lambert/dynamic.zip*

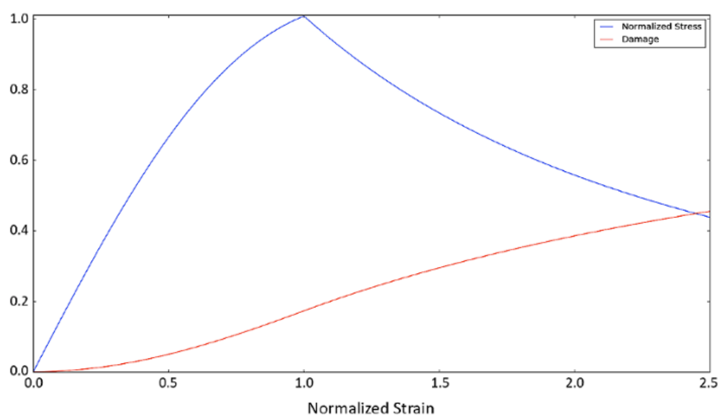

Figure 3.9. *Evolution of stress and damage for a 1D ductile material (using the von-Mises criterion) for the case $E(\alpha)=\sigma_Y(\alpha)=(1-\alpha)^2$ and $w(\alpha)=\alpha^2$. For a color version of the figure, see www.iste.co.uk/lambert/dynamic.zip*

The study of the damage evolution consists of defining an energy and finding its minimum with respect to α:

$$\mathcal{E}(u,\alpha) = \int_\Omega \Big(\frac{1}{2}E_0 a(\alpha)\big(\varepsilon(u)\big)^2 + w_1\alpha + \frac{1}{2}w_1\ell^2(\alpha')^2\Big)d\Omega \qquad [3.111]$$

The dynamic equation in 1D can be written as

$$\rho \ddot{u} = \sigma' = E_0\big(a(\alpha)\varepsilon(u)\big)' \qquad [3.112]$$

and we impose a displacement on the extremities:

$$\begin{cases} u(x=0,t) = 0 \\ u(x=L,t) = \dot{\varepsilon}_0 L t \end{cases} \qquad [3.113]$$

For this model, we have the following parameters: L, E_0, w_1, ℓ, ρ and $\dot{\varepsilon}_0$.

The first step is to reduce the number of parameters of the problem.

Since we are interested in finding the minimizer of $\mathcal{E}$, it is clear that we can redefine $\mathcal{E}$ as

$$\mathcal{E}(u,\alpha) = \int_\Omega \Big(\frac{1}{2}\frac{E_0}{w_1}a(\alpha)\big(\varepsilon(u)\big)^2 + \alpha + \frac{1}{2}\ell^2(\alpha')^2\Big) d\Omega \qquad [3.114]$$

It is clear that the dynamic equation depends only on $\frac{E_0}{\rho}$. Therefore, we are only interested in five values: L, $\frac{E_0}{w_1}$, $\frac{E_0}{\rho}$, ℓ and $\dot{\varepsilon}_0$.

We will change the scale of our variables in order to remove three parameters from our problem.

We first write $\tilde{x} = \frac{1}{L}x$ and $\tilde{t} = Tt$, for some constant $T > 0$ that we will specify later.

If $x \in \Omega$, then $\tilde{x} \in [0,1]$. The imposed displacements are now

$$\begin{cases} u(\tilde{x}=0,t) = 0 \\ u(\tilde{x}=1,t) = \dot{\varepsilon}_0 L t \end{cases} \qquad [3.115]$$

If $f(x,t)$ is a function of x and t, then we define

$$\tilde{f}(\tilde{x},\tilde{t}) := f(x,t) \qquad [3.116]$$

We derive it to obtain

$$\frac{df(x,t)}{dx} = \frac{1}{L}\frac{d\tilde{f}(\tilde{x},\tilde{t})}{d\tilde{x}} \qquad \text{and} \qquad \frac{df(x,t)}{dt} = \frac{1}{T}\frac{d\tilde{f}(\tilde{t},\tilde{t})}{d\tilde{t}} \qquad [3.117]$$

Suppose we have a constant $U_0 > 0$. We define

$$\begin{cases} \tilde{u}(\tilde{x}, \tilde{t}) := \frac{1}{U_0} u(x,t) \\ \tilde{\alpha}(\tilde{x}, \tilde{t}) := \alpha(x,t) \end{cases} \tag{3.118}$$

Thus

$$\mathcal{E}(\tilde{u}, \tilde{\alpha}) = \int_\Omega \Big(\frac{1}{2} \frac{E_0 U_0^2}{w_1 L^2} a(\tilde{\alpha}) \big(\frac{d\tilde{u}}{d\tilde{x}} \big)^2 + \tilde{\alpha} + \frac{1}{2} \frac{\ell^2}{L^2} \big(\frac{d\tilde{\alpha}}{d\tilde{x}} \big)^2 \Big) d\Omega \tag{3.119}$$

and

$$\frac{1}{T^2} \frac{d^2 \tilde{u}}{d\tilde{t}^2} = \frac{E_0}{\rho L^2} \frac{d}{d\tilde{x}} \big(a(\tilde{\alpha}) \frac{d\tilde{u}}{d\tilde{x}} \big) \tag{3.120}$$

Since we only assumed T and U_0 as two positive constants, we can now fix them. We set $U_0 = L\sqrt{\frac{w_1}{E_0}}$ and $T = L\sqrt{\frac{\rho}{E_0}}$. We also define $\tilde{\ell} := \frac{\ell}{L}$ and $\tilde{\dot{\varepsilon}}_0 := \dot{\varepsilon}_0 \frac{TL}{U_0}$.

We have

$$\mathcal{E}(\tilde{u}, \tilde{\alpha}) = \int_\Omega \Big(\frac{1}{2} a(\tilde{\alpha}) \big(\frac{d\tilde{u}}{d\tilde{x}} \big)^2 + \tilde{\alpha} + \frac{1}{2} \tilde{\ell}^2 \big(\frac{d\tilde{\alpha}}{d\tilde{x}} \big)^2 \Big) d\Omega \tag{3.121}$$

and the dynamics of the system is

$$\frac{d^2 \tilde{u}}{d\tilde{t}^2} = \frac{d}{d\tilde{x}} \big(a(\tilde{\alpha}) \frac{d\tilde{u}}{d\tilde{x}} \big) \tag{3.122}$$

Considering $\tilde{x}$ and $\tilde{t}$ as the space and time variables (and removing the tilde from our notation), we obtain the dimensionless problem in $\Omega = [0, 1]$:

– The damage profile α minimizes the energy $\mathcal{E}$ taking into account the irreversibility condition, where $\mathcal{E}$ is given by

$$\mathcal{E}(u, \alpha) = \int_\Omega \Big(\frac{1}{2} a(\alpha) \big(\varepsilon(u) \big)^2 + \alpha + \frac{1}{2} \ell^2 \big(\alpha' \big)^2 \Big) d\Omega \tag{3.123}$$

– The time evolution of the displacement u is given by

$$\ddot{u} = \big(a(\alpha) \varepsilon(u) \big)', \tag{3.124}$$

under the imposed boundary conditions

$$\begin{cases} u(x{=}0,t) = 0 \\ u(x{=}1,t) = \dot{\varepsilon}_0 t \end{cases} \qquad [3.125]$$

EXAMPLE 3.5.– *Suppose we want to study the fracture of a bar made of steel. We assume that this bar is 10 cm long and is being stretched at a constant speed of 100 m/s. For this material, we have a density ρ = 8,000 kg/m^3, modulus of elasticity $E_0 = 210$ GPa, fracture toughness $K_{IC} = 50$ MPa$\cdot m^{1/2}$ and ultimate tensile strength σ_c = 1,000 MPa.*

When using this damage gradient model in a quasi-static scenario, we know that

$$w_1 = \frac{\sigma_C^2}{E_0} = \frac{(1000 \cdot 10^6)^2}{210 \cdot 10^9} Pa = 4.7\ MPa$$

The fracture energy G_c rate can be found as

$$G_c = \frac{K_{IC}^2}{E_0} = \frac{(50 \cdot 10^6)^2}{210 \cdot 10^9} = 11.7\ kPa \cdot m.$$

The energy dissipated by the fracturing process, given by equation [3.35], can be written as

$$G_c = \ell \sigma_C \frac{4\sqrt{2}}{3}$$

We obtain $\ell = 1.2 \cdot 10^{-5}$ m and $\tilde{\ell} = \ell / L = 1.2 \cdot 10^{-4}$.

The values of T and U_0 are

$$T = L\sqrt{\frac{\rho}{E_0}} = 0.1\sqrt{\frac{8000}{210 \cdot 10^9}} s = 2 \cdot 10^{-5} s$$

$$U_0 = L\sqrt{\frac{w_1}{E_0}} = 0.1\sqrt{\frac{4.7 \cdot 10^6}{210 \cdot 10^9}} m = 4.7 \cdot 10^{-4} m.$$

The deformation can be now be found as

$$\varepsilon = \frac{du}{dx} = \frac{U_0}{L}\frac{d\tilde{u}}{d\tilde{x}} = 4.73 \cdot 10^{-3}\frac{d\tilde{u}}{d\tilde{x}}.$$

The dimensionless deformation speed is

$$\tilde{\dot{\varepsilon}}_0 = \dot{\varepsilon}_0 \frac{TL}{U_0} = 4.3 \cdot 10^{-3} \dot{\varepsilon}_0 = 0.43$$

This bar can be simulated using our model with only two parameters ($\tilde{\ell} = 1.2 \cdot 10^{-4}$ and $\tilde{\dot{\varepsilon}}_0 = 0.43$).

When analyzing the results, it must be kept in mind that a time of 1 in the simulation is equivalent to $2 \cdot 10^{-5}$ s. In the same way, a deformation of 1 in the simulation is equivalent to $\varepsilon = 4.73 \cdot 10^{-3}$ in the real bar.

3.4.4. *1D period bar*

We are interested in obtaining the number of fragments of a ring under expansion. Instead of working with a ring in a 3D scenario, we consider a bar $[0, L]$ and the following periodic conditions in the strain ε and damage α:

$$\begin{cases} \varepsilon(x+L, t) = \varepsilon(x, t), \, x \in \mathbb{R} \\ \alpha(x+L, t) = \alpha(x, t), \, x \in \mathbb{R}, \end{cases} \qquad [3.126]$$

for every $t \in \mathbb{R}$.

We also suppose that we start our study with a completely healthy bar under a uniform (in space) strain rate $\dot{\varepsilon}_0$. At the instant $t = 0$, we have

$$\begin{cases} \dot{\varepsilon}(x, 0) = \dot{\varepsilon}_0, \, x \in [0, L] \\ \alpha(x, 0) = 0, \, x \in [0, L] \end{cases} \qquad [3.127]$$

For simplicity, we assume that the initial strain is zero, i.e. $\varepsilon(x, 0) = 0, x \in [0, L]$. The ring of initial perimeter L suffers uniform expansion. At an instant t, the perimeter of the ring is $L+\dot{\varepsilon}_0 Lt$. The displacement u at the extremities and ε must satisfy

$$\int_0^L \varepsilon(x, t) dx = u(L, t) - u(0, t) = \dot{\varepsilon}_0 Lt \qquad [3.128]$$

It is clear that $\varepsilon(x, t) = \dot{\varepsilon}_0 t$ satisfies the above equations.

At each instant of time t, we write

$$\varepsilon(x,t) = \varepsilon^*(x,t) + \dot{\varepsilon}_0 t \qquad [3.129]$$

The variable ε^* is the difference between the real strain and uniform strain. It is easy to see that ε^* is periodic in x and

$$\int_0^L \varepsilon^*(x,t) = 0 \qquad [3.130]$$

We define the function u^* as the difference between the real displacement and uniform displacement, i.e.

$$u^*(x,t) = u(x,t) - \dot{\varepsilon}_0 x t \qquad [3.131]$$

By differentiating the above equation, we find that $(u^*)' = \varepsilon^*$. It is clear that u^* is periodic.

The stress can be written as

$$\sigma(x,t) = A(\alpha)((u^*)' + \dot{\varepsilon}_0 t) \qquad [3.132]$$

The dynamic equation is

$$\ddot{u}^*(x,t) = \ddot{u}(x,t) = \sigma' \qquad [3.133]$$

Finally, we write the total energy as

$$\mathcal{E}(u^*,\alpha) = \int_0^L \frac{1}{2}A(\alpha)((u^*)' + \dot{\varepsilon}_0 t)^2 + w(\alpha) + \frac{1}{2}w_1\ell^2(\alpha')^2 \qquad [3.134]$$

The problem consists of finding two periodic variables u^* and α satisfying the dynamic equation and $\frac{d\mathcal{E}}{d\alpha}(u^*,\alpha)\beta = 0$ for every β admissible.

3.4.4.1. *Influence of each parameter on damage*

We now want to investigate the influence of the parameters on the number of cracks in the bar. As we have seen, the gradient damage model can be described by two parameters: the characteristic length ℓ and the deformation speed $\dot{\varepsilon}_0$.

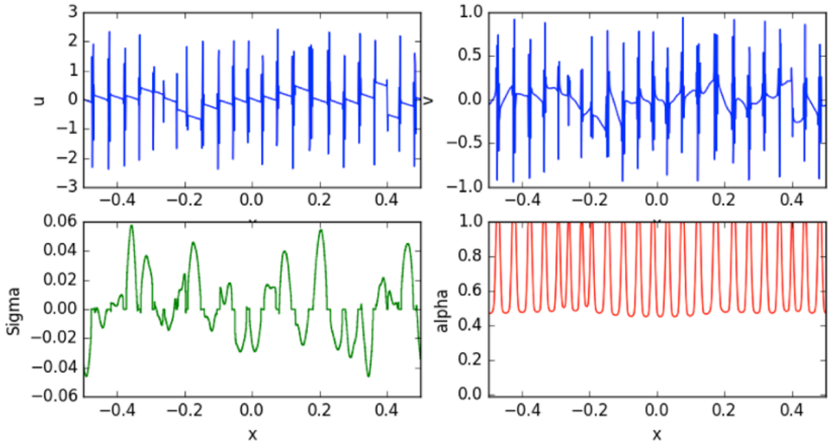

Figure 3.10. *The displacement u^* is at the top left and the damage profile is at the bottom right. We can also see the velocity distribution and stress σ. For a color version of the figure, see www.iste.co.uk/lambert/dynamic.zip*

There is, however, another parameter we need to pay attention to: the size of elements Δx. Even though it is a purely numerical parameter, it has great importance in the number of fragments we are able to obtain. Therefore, for each pair $(\ell, \dot{\varepsilon}_0)$ we want to study, we run several simulations with different values of Δx and see if the number of fragments converge for $\Delta x \to 0$. We emphasize that we are interested in the convergence of the number of fragments and not in the convergence of u and α. Since we have a periodic problem, a translation of (u, α) would be a different numerical result, but the same physical result.

Numerical simulations show that there is only a small difference between the results if the mesh is fine enough. The number of fragments seem to converge as $\Delta x \to 0$.

In Figure 3.11, we can see the influence of the mesh and the number of elements used. We recall that we used standard P1 elements. We note that for $1/\ell \leq$ 1,000, a simulation using 2,000 elements (N_{elem} =2,000) is accurate. For $1/\ell \leq$ 2,000, 5,000 elements is enough. This holds true for other values of ℓ and N_{elem}. Hence, we will consider that the results are accurate if $1/\ell \leq \frac{1}{3} N_{elem}$ or, equivalently, $\ell \geq 3/N_{elem}$. For other values of $\dot{\varepsilon}_0$, this relation also seems to be valid.

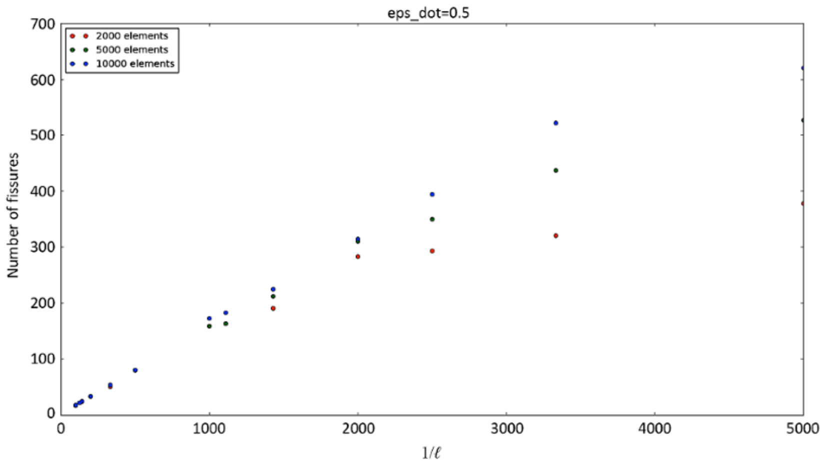

Figure 3.11. *We see that the number of cracks is almost proportional to $1/\ell$ if the mesh is fine enough; here, $\dot{\varepsilon}_0 = 0.5$. For a color version of the figure, see www.iste.co.uk/lambert/dynamic.zip*

Again from Figure 3.11, we can see that the number of fragments increases linearly with $1/\ell$ for $\dot{\varepsilon}_0 = 0.5$. This linear behavior holds true for every other value of $\dot{\varepsilon}_0$ between 10^{-4} and 10^2 tested.

We can see the influence of $\dot{\varepsilon}_0$ in Figure 3.12. We make $\dot{\varepsilon}_0$ vary between 10^{-4} and 10^2. The number of cracks change from less than 500 to more than 1,000.

We show in Figure 3.13 the results in a log–log scale. When $\dot{\varepsilon}_0 \geq 1$, the points appear to be in a straight line.

We conclude with two remarks. First, we can see that the behavior is not monotonic. For close values of $\dot{\varepsilon}_0$, we see that there is an oscillation in the number of fragments. Nevertheless, we see a clear tendency for the number of fragments to increase as the strain rate increases.

Second, we remark that a small initial perturbation does not change the number of cracks in the end. There is, however, some changes in how they appear and how many of these cracks develop until total failure, but we emphasize that the number of cracks in the end is the same if the perturbation is sufficiently small.

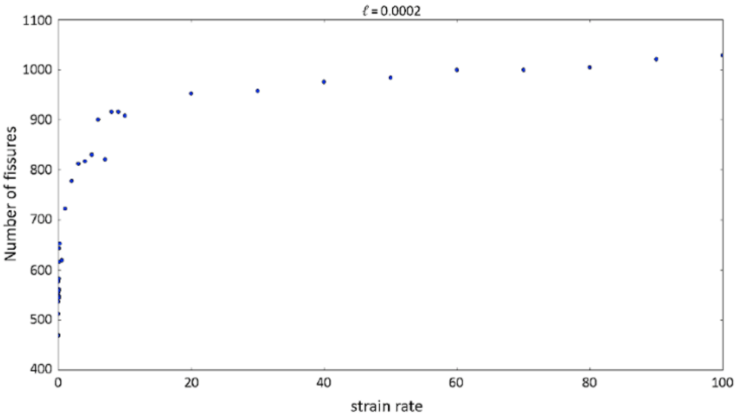

Figure 3.12. *Influence of $\dot{\varepsilon}_0$ on the number of cracks for ℓ=$2 \cdot 10^{-4}$*

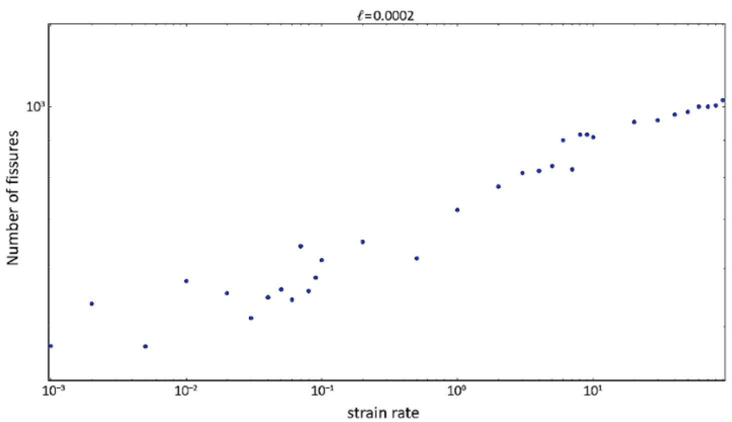

Figure 3.13. *Influence of $\dot{\varepsilon}_0$ for a fixed value of ℓ (log–log)*

3.4.5. *Cylinder under internal pressure*

The final application concerns the study of the fragmentation of a cylinder under a strong internal pressure. We want to know when and how the material breaks. In this case, we expect the cylinder to fragment into multiple parts.

The cylinder has an internal radius of R_i=1.0 and an external radius of R_e=1.25. We impose an internal pressure of 1.0 and we assume it to be constant in space and time. The following material constants are used: density, ρ=10^3; Young's modulus, E_0=10^4 and Poisson coefficient, ν=0.3.

We first study a brittle cylinder. We consider the dependency of the rigidity tensor with respect to α as $E(\alpha)=\mathbf{E}_0(1-\alpha)^2$, which we have already studied (Figure 3.1). The damage coefficients used in the energy are $w_1=8{\cdot}10^{-3}$ and the characteristic length $\ell=10^{-2}$. For these parameters, the critical stress is $\sigma_c=\sqrt{E_0 w_1}=8.9$, which is much larger than the pressure imposed. However, the internal pressure causes waves of stress to propagate and, by combining with each other, they reach the stress threshold and cause fracture on the cylinder.

The problem in question is axisymmetric and we could expect a radial symmetry of the results. In practice, a small perturbation is enough to localize damage. One could argue that by changing the perturbation used, we could obtain different results. This is true; however, we are interested in the moment of fragmentation and the number of fragments, which do not seem to change when the perturbation is sufficiently small.

We see the damage profile in Figure 3.14. We can see the cracks in red and the healthy material in blue. There are two main aspects that we see in this figure: the first one is that we have several damage bands and that they are somewhat evenly distributed. The second is the direction of the cracks. We see that the cracks follow the radial direction almost everywhere.

Next, we consider a ductile cylinder. We consider once again $E(\alpha)=\mathbf{E}_0(1-\alpha)^2$ and, now, $\sigma_Y(\alpha)=\sigma_Y^0(1-\alpha)^2$ (see Figure 3.4). We take the initial yield stress as $\sigma_Y^0=5.0$, such that it is greater than the applied pressure but smaller than the critical damage stress. In this way, we obtain an elastic phase, a phase where plastic strain occurs but not damage and a phase with damage and plastic strain evolution.

The damage profile in Figure 3.15 is different from the one obtained for brittle materials. We can see once again that we have damage bands that are evenly distributed, but their direction has changed. As expected for ductile materials, when combining damage and plasticity, the cracks evolve in a $45°$ angle.

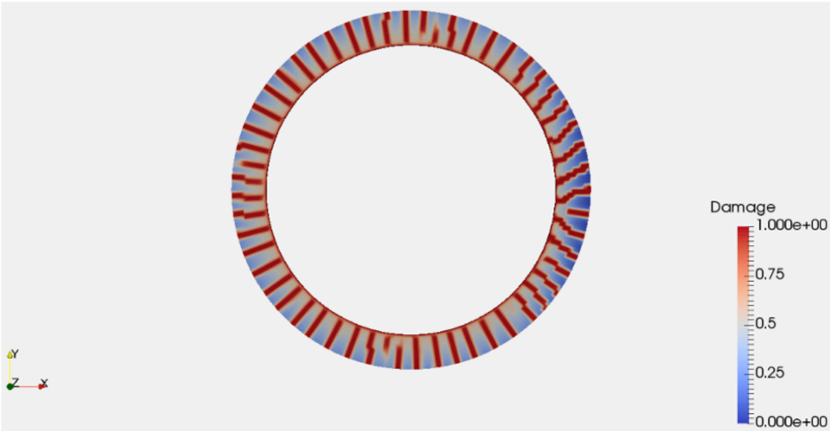

Figure 3.14. *Damage profile for a brittle cylinder under internal pressure. We consider $E(\alpha) = \mathbf{E}_0(1-\alpha)^2$ and $w(\alpha) = w_1\alpha$. For a color version of the figure, see www.iste.co.uk/lambert/dynamic.zip*

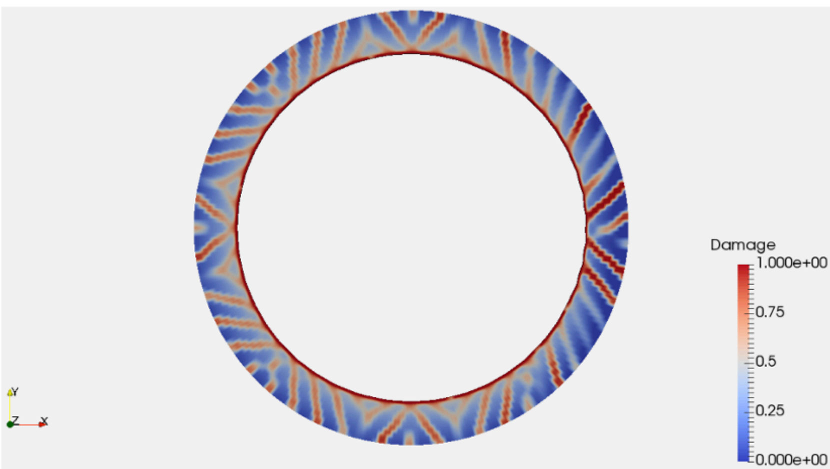

Figure 3.15. *Damage profile for a ductile cylinder under internal pressure. We consider $E(\alpha) = \mathbf{E}_0(1-\alpha)^2$, $\sigma_Y(\alpha) = \sigma_Y^0(1-\alpha)^2$ and $w(\alpha) = w_1\alpha$. For a color version of the figure, see www.iste.co.uk/lambert/dynamic.zip*

The last question we want to discuss is the number of fragments. We recall that the characteristic length is proportional to the thickness of the damage band for a quasi-static loading. For a dynamic problem, this is no longer true, but it is still the most important parameter for determining the number of cracks. If we decrease the characteristic length, since each crack occupies less space, we should obtain more cracks.

To test this hypothesis, we run a simulation with a characteristic length 10 times smaller ($\ell = 10^{-3}$). The result (Figure 3.16) is very clear. We can see that each crack is now thinner and we have many more cracks. We also note that there are now many cracks that cross each other.

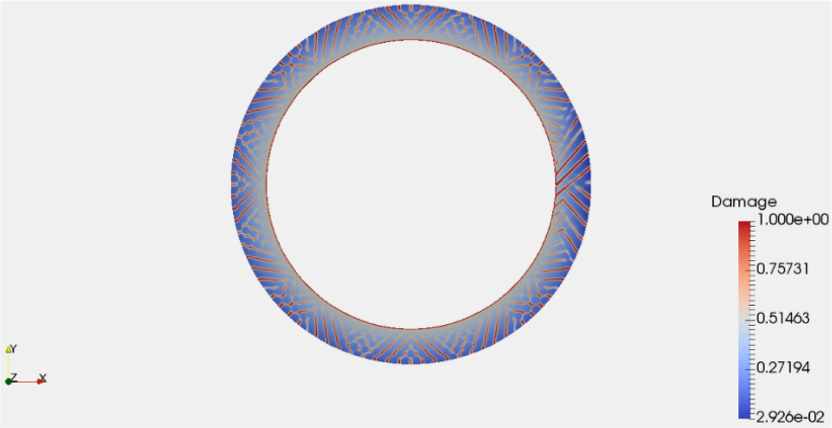

Figure 3.16. *Damage profile for a ductile material and smaller characteristic length. For a color version of the figure, see www.iste.co.uk/lambert/dynamic.zip*

3.5. Conclusion

In this work, we have briefly explained the hypotheses considered in the construction of gradient damage models for brittle softening materials for an infinitely slow loading, based on the principle of minimum energy. We then explained the necessary changes to take plastic strains and inertia into account.

We have shown that these models are very flexible, and only by changing the dependence of the rigidity and yield stress on the damage can we model material behaviors that are very different.

We then concluded by some applications, showing the influence of each parameter of the model and by studying the fragmentation of a cylinder under initial pressure.

3.6. References

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